Nettverk og systemadministrasjon (DAVE3610)

Weekly assignments

By Jørgen Skontorp

The solution for these assignments are based on demos provided by Ismail Hassan.

Licensed under the CreativeCommons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license.

Disclaimer:

- I do not vouch for my solutions. Follow at your own risk.
- The IP addresses, passwords, etc. are all tied to my personal OsloMet server and are just for me. If you follow this guide, you need to set your own.
- I use Neovim as my default text editor. If you use anything else, like nano, switch out nvim with nano.

Assignment-1: Practice working with directories....3
Assignment-2: Disks, partitions and file systems....5
Assignment-3: Managing local storage (RAID and LVM)....11
Step 1: Clone your virtual machine....12
Step 2: Attach 4 extra disks to the cloned virtual machine....13
Step 3: Checking and preparing the disks for RAID....14
Step 4: Install the necessary packages....17
Step 5: Create a RAID Volume....18
Step 6: Creating Logical Volumes....21
Step 7: Creating filesystem on the Logical Volumes....24
Step 8: Simulating a failed disk....27
Step 9: Resizing LVM and the filesystem....30
Assignment-4: Software Installation (dpkg, apt and from source)....32
Step 1: Installing software without dependencies with dpkg....33
Step 2: Installing software with dependencies with dpkg....35
Step 3: Exploring the apt package management tool used by Ubuntu....37
Step 4: Installing software from source....39
Step 5: Installing software with checkinstall....41
Assignment-5: User management (Local Accounts)....43
Step 1: Get information about the users on the system....44
Step 2: Create the groups....46

Step 3: Create the users....47
Step 4: Creating users in a bulk....48
Step 5: Account management....51
Assignment-6: Centralized Account Management and Quotas....52
Step 1: Provision instances on OpenStack Cloud....52
Step 2: Setup the LDAP server and the LDAP client....53
Assignment-7: Managing permission....68
Step 1: Managing Permissions:....69
Chapter 32....73
Chapter 33....76
Step 2 - Managing quotas....78
Step 3 - Delegating root privileges through sudo....81
Assignment-8: Centralized Home Directories through NFS....82
Assignment-9: Storage Area Network (SAN)....86
Step 1: Preparing the machines needed for an iSCSI server and client....86
Step 2: Setup the iSCSI server (Target)....86
Step 3: Install and configure iSCSI client (Initiator)....91
Assignment-10.1: Dynamic Host Configuration Protocol (DHCP)....94
Assignment-10.2: Domain Name Server (DNS)....98
Assignment-11: Email system....102
MTA/SMTP....103
IMAP with SSL/TLS....109
Webmail....113
SMTP-AUTH....115
Spam and virus filter....117
Multiple domains....120

Assignment-1: Practice working with directories

1. Display your current directory.

```
ubuntu@first-instance:~$ pwd
```

2. Change to the /etc directory.

```
ubuntu@first-instance:~$ cd /etc
```

3. Now change to your home directory using only three key presses.

```
ubuntu@first-instance:~$ cd
```

4. Change to the /boot/grub directory using only eleven key presses.

```
ubuntu@first-instance:~$ cd /boot/grub
```

5. Go to the parent directory of the current directory.

```
ubuntu@first-instance:~$ cd ..
```

6. Go to the root directory.

```
ubuntu@first-instance:~$ cd /
```

7. List the contents of the root directory.

```
ubuntu@first-instance:~$ ls /
```

8. List a long listing of the root directory.

```
ubuntu@first-instance:~$ ls -l /
```

9. Stay where you are, and list the contents of /etc.

```
ubuntu@first-instance:~$ ls /etc
```

10. Stay where you are, and list the contents of /bin and /sbin.

```
ubuntu@first-instance:~$ ls /bin /sbin
```

11. Stay where you are, and list the contents of ~.

```
ubuntu@first-instance:~$ ls ~
```

12. List all the files (including hidden files) in your home directory.

```
ubuntu@first-instance:~$ ls -a ~
```

13. List the files in /boot in a human readable format.

```
ubuntu@first-instance:~$ ls -lh /boot
```

14. Create a directory testdir in your home directory.

```
ubuntu@first-instance:~$ mkdir ~/testdir
```

15. Change to the /etc directory, stay here and create a directory newdir in your home directory.

```
ubuntu@first-instance:~$ cd /etc
ubuntu@first-instance:~$ mkdir ~/newdir
```

16. Create in one command the directories ~/dir1/dir2/dir3 (dir3 is a subdirectory from dir2, and dir2 is a subdirectory from dir1)

```
ubuntu@first-instance:~$ mkdir -p ~/dir1/dir2/dir3
```

17. Remove the directory testdir.

```
ubuntu@first-instance:~$ rm -r ~/testdir
```

18. If time permits (or if you are waiting for other students to finish this practice), use and understand pushd and popd. Use the man page of bash to find information about these commands.

```
ubuntu@first-instance:~$ pushd /var # Saves current directory and changes to /var
ubuntu@first-instance:~$ popd # Pops top directory from the stack and changes back to it
```

Assignment-2: Disks, partitions and file systems

The goal of this assignment is to get accustomed to how to:

- locate and recognise hard disk devices in Linux.
- create partitions on those devices.
- create/place a file system on the created partitions.
- mount the file system.

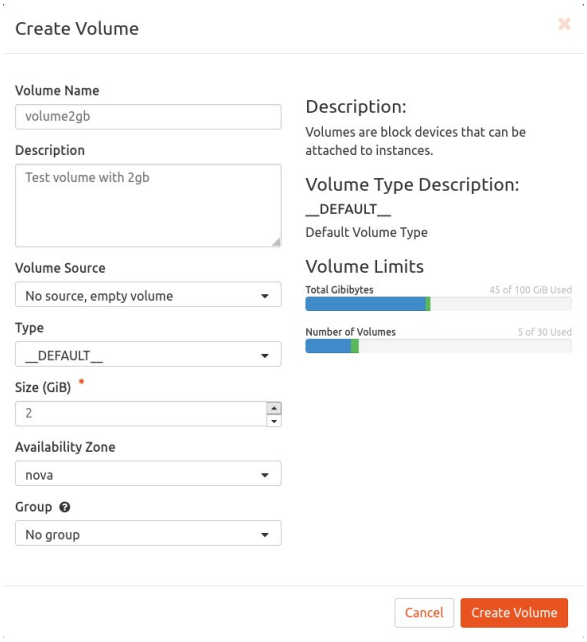
As the newly recruited system administrator you receive a mail from the senior system administrator asking you to carry out the following:

Hi Novice Sysadmin,

Great work last week on getting accustomed to the Virtualization platform that our company uses. It is essential that you get very familiar with it. As you will be doing a lot of installations, setups and configurations, it is essential that you document your work that other sysadmin would be able to replicate your work.

We would like you to attach 2 new disks (a **3GB** disk and a **2GB** disk) to the image/machine you imported to VirtualBox last week. Start the machine and do the following:

I created 2 volumes in Caonical OpenStack by going to Volumes:



Then manage attachments and add to first_instance

- **Locate and recognise hard disk devices:**
  - Verify that all the new disks are properly detected at boot-up.
  - Verify that you can see the disk devices in /dev.
  - Use *fdisk* to display the total size of the new disks.

```
ubuntu@first-instance:~$ sudo fdisk --list
Disk /dev/loop0: 63.91 MiB, 67014656 bytes, 130888 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop1: 114.41 MiB, 119967744 bytes, 234312 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 111.95 MiB, 117387264 bytes, 229272 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop3: 40.84 MiB, 42827776 bytes, 83648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop4: 63.46 MiB, 66547712 bytes, 129976 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop5: 40.43 MiB, 42393600 bytes, 82800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 8A49268C-6948-4D42-A28B-C234361988B6

Device       Start      End  Sectors  Size Type
/dev/sda1   227328 20971486 20744159  9.9G Linux filesystem
/dev/sda14    2048    10239     8192    4M BIOS boot
/dev/sda15   10240   227327   217088  106M EFI System

Partition table entries are not in disk order.

Disk /dev/sdb: 2 GiB, 2147483648 bytes, 4194304 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdc: 3 GiB, 3221225472 bytes, 6291456 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

- **Create partitions on the new devices:**
  - Create a 200MB primary partition on the small disk.
  - Create a 400MB primary partition and two 300MB logical drives on the big disk.
  - Use **df -h** and **fdisk -l** to verify the partitions.
  - Take a backup of the partition table containing your 400MB primary and 300MB logical drives. Make sure the logical drives are in the backup.
  - Take a backup of the partition table containing your 200MB primary partition.
  - **(optional)** Remove all your partitions with **fdisk**. Then restore your backups.

```
# Create 200MB partition on sdb
ubuntu@first-instance:~$ sudo fdisk /dev/sdb # With m, n, p, 1, +200MB, w

# Create 400MB primary, extended container and two 300MB logical disks
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # with m, n, p, 1 +400MB
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # with m, n, e, +650M
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # with m, n, l, 5 +300MB
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # with m, n, l, 6 +300MB

# Check the partition
ubuntu@first-instance:~$ sudo fdisk -l

# Create backup of partitions
ubuntu@first-instance:~$ sudo fdisk /dev/sdb # O small_disk_backup
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # O big_disk_backup

# Delete all partitions
ubuntu@first-instance:~$ sudo fdisk /dev/sdb # with d
ubuntu@first-instance:~$ sudo fdisk /dev/sdc # with d three times

# Import partition backups
ubuntu@first-instance:~$ sudo fdisk dev/sdb # I small_disk_backup
ubuntu@first-instance:~$ sudo fdisk dev/sdc # I big_disk_backup

# Check the partition
ubuntu@first-instance:~$ sudo fdisk -l
```

- **Create/place a file system on the created partitions:**
  - Create an ext2 filesystem on the 200MB partition.
  - Create an ext3 filesystem on one of the 300MB logical drives.
  - Create an ext4 on the 400MB partition.
  - **(optional)** Create an XFS filesystem on one of the other 300MB logical drives.
  - Verify your work with **fdisk** and **df**.
  - Perform a file system check on the new file systems. (Only on ext2, ext3 and ext4)

```
# Check the partitions
ubuntu@first-instance:~$ sudo blkid | grep sd

# Create ext2 filesystem on the 200MB partition
ubuntu@first-instance:~$ sudo mkfs -t ext2 /dev/sdb1

# Create ext3 filesystem on one logical drive
ubuntu@first-instance:~$ sudo mkfs -t ext3 /dev/sdc5

# Create ext4 on the 400MB partition
ubuntu@first-instance:~$ sudo mkfs -t ext4 /dev/sdc1

# Create XFS filesystem on the other logical drive
ubuntu@first-instance:~$ sudo mkfs.xfs /dev/sdc6

# Check the filesystem on the partitions
ubuntu@first-instance:~$ sudo blkid | grep sd

# Perform file system check on the ext file systems
ubuntu@first-instance:~$ sudo fsck /dev/sdb1
ubuntu@first-instance:~$ sudo fsck /dev/sdc5
ubuntu@first-instance:~$ sudo fsck /dev/sdc1
```

- **Mount the file system:**
  - Create the following directories:
    - /Dave3610/ext2
    - /Dave3610/ext3
    - /Dave3610/ext4
  - Mount the small 200MB partition on /Dave3610/ext2
  - Mount the 300MB partition with the ext3 filesystem on /Dave3610/ext3
  - Mount the 400MB partition on /Dave3610/ext4
  - Verify your work with **fdisk**, **df** and **mount**. Also look in /etc/mtab and /proc/mounts
  - Make all the mounts permanent, test that it works.
    - The mounts should be available after system reboot!
  - **What happens when you mount a file system on a directory that contains some files?**
  - **What happens when you mount two file systems on the same mount point?**

```
# Create root directories for mounting
ubuntu@first-instance:~$ sudo mkdir -p /Dave3610/ext2
ubuntu@first-instance:~$ sudo mkdir -p /Dave3610/ext3
ubuntu@first-instance:~$ sudo mkdir -p /Dave3610/ext4

# Check directories
ubuntu@first-instance:~$ ls -l /Dave3610/

# Check what is already mounted
ubuntu@first-instance:~$ mount | grep sd # Only sda1 and boot mounted

# Mount the 200MB partition (ext2) on Dave3610/ext2
ubuntu@first-instance:~$ sudo mount -t ext2 /dev/sdb1 /Dave3610/ext2

# Mount the 300MB partition (ext3) on /Dave3610/ext3
ubuntu@first-instance:~$ sudo mount -t ext3 /dev/sdc5 /Dave3610/ext3

# Mount the 400MB partition on /Dave3610/ext4
ubuntu@first-instance:~$ sudo mount -t ext4 /dev/sdc1 /Dave3610/ext4

# Verify with fdisk, df and mount
ubuntu@first-instance:~$ mount | grep sd
# sdb1, sdc5 and sdc1 mounted correctly
ubuntu@first-instance:~$ df
# sdb1, sdc5 and sdc1 mounted correctly
ubuntu@first-instance:~$ sudo fdisk -l # All partitions present

# Look in /etc/mtab and /proc/mounts
ubuntu@first-instance:~$ cat /etc/mtab
ubuntu@first-instance:~$ cat /proc/mounts
/dev/sdb1 /Dave3610/ext2 ext2 rw,relatime 0 0
/dev/sdc5 /Dave3610/ext3 ext3 rw,relatime 0 0
/dev/sdc1 /Dave3610/ext4 ext4 rw,relatime 0 0

# Make mounts permanent
ubuntu@first-instance:~$ sudo nvim /etc/fstab
# Add the lines from /etc/mtab above then :!wq

# Check mount
ubuntu@first-instance:~$ mount -a
```

```
# No result, so everything seeHeading 1ms ok


# Reboot to check if mount is permanent
ubuntu@first-instance:~$ sudo reboot
ubuntu@first-instance:~$ ssh ubuntu@ip
ubuntu@first-instance:~$ mount | grep sd
# All filesystems are mounted at the right paths

# Try mount on directory which contains files
ubuntu@first-instance:~$ mkdir testmount
ubuntu@first-instance:~$ mv *.txt ./testmount
ubuntu@first-instance:~$ sudo mount -t xfs /dev/sdc6 /home/ubuntu/testmount
ubuntu@first-instance:~$ ls testmount/
# All previous files are gone

# Try to mount two filesystems on the same mount point
ubuntu@first-instance:~$ sudo mount -t ext2 /dev/sdb1 /home/ubuntu/testmount
ubuntu@first-instance:~$ sudo mount -t ext3 /dev/sdc5 /home/ubuntu/testmount
ubuntu@first-instance:~$ mount
/dev/sdc6 on /home/ubuntu/testmount type xfs
(rw,relatime,attr2,inode64,logbufs=8,logbsize=32k,noquota)
/dev/sdb1 on /home/ubuntu/testmount type ext2 (rw,relatime)
/dev/sdc5 on /home/ubuntu/testmount type ext3 (rw,relatime)

# Found out that I have to mount with UUID and nofail,auto to avoid crash at
startup
ubuntu@first-instance:~$ blkid
ubuntu@first-instance:~$ sudo nvim /etc/fstab
UUID='372d5496-a3fc-4dd2-b4dc-4ade435ba404' /Dave3610/ext2 ext2
rw,relatime,nofail,auto 0 0
UUID='c3cbb48a-8253-4864-a60f-318c07b9dbf9' /Dave3610/ext3 ext3
rw,relatime,nofail,auto 0 0
UUID='131f1ce9-913a-4ca3-9f28-6ac173fede60' /Dave3610/ext4 ext4
rw,relatime,nofail,auto 0 0

# When mounting a filesystem to a directory containing files, the files will be
“hidden behind” the mounted filesystem while it is mounted. If you mount two
filesystems on the same mount point, all files will be “hidden behind” the last
filesystem mounted until you unmount.
```

Assignment-3: Managing local storage (RAID and LVM)

The goal of this week's assignment is to:

- understand the partitioning scheme of Linux systems.
- become accustomed to Software Raid technology.
- understand how to utilize Logical Volume management.
- understand how to utilize filesystems on top of LVM and RAID.

As the newly recruited system administrator you receive the following mail from the senior system administrator:

Hi Novice Sysadmin,

we have just purchased a new server with a lot of disk space but due to financial constraints, we, unfortunately, did not order or include any Hardware Raid controllers.

Disk failures are very common in the datacenter and in order to mitigate the problem, we have opted for the use of **Linux Software Raid** technology. We would in addition like to avoid the less flexible traditional partitioning scheme that we have been using so far and rather utilize the power and flexibility of Linux **Logical Volume Management (LVM)** technology.

Before we implement both technologies on the new server we would like you to try it out on the virtual machine you worked on last week and document the procedure. Your task is to do the following steps on the next page:

Best regards
Senior Sysadmin!

Step 1: Clone your virtual machine

Since you will be doing quite a lot of hands-on tasks, it is often wise to work on a fresh image. This way, you will not be hindered by mistakes or misconfigurations introduced in previous steps or tasks. The beauty of virtual machines is that you can easily dispose of them and recreate/clone them from your original image at a very fast pace.

- Turn off the Virtual Machine.
- Clone the Virtual Machine.

```
Since we are not using a virtual machine, I am creating a snapshot. A snapshot
must be of a running machine.

Start instance > create snapshot > «lecture 2» > save
```

Step 2: Attach 4 extra disks to the cloned virtual machine

The following steps should be done on the newly **cloned** machine.

- Add/attach **4** disks (**HD-1**, **HD-2**, **HD-3** and **HD-4**) and give each disk a capacity of **1 GB**.

```
In the cloud panel:
# Create new volumes
Volumes > Volumes
+ Create volume
Create HD-1, HD-2, HD-3 and HD-4 with 1GB capacity with default settings
# Add volumes to instance
Edit volume > Manage attatchments > Select an instance > first_instance >
attatch
Disks added to /dev/sdd, /dev/sde, /dev/sdf and /dev/sdg
```

Step 3: Checking and preparing the disks for RAID

- Start and login to the machine
- Become root with the **sudo** command
- Check the new disks with the **fdisk -l** command:
- Create a new primary partition on each of the newly attached disks av type **Linux raid auto** that spans over the whole disk. In other words, the disk should only have 1 primary partition that uses the whole disk.
- Check the new partitions with the **fdisk -l** command and make sure that they are of type **Linux raid auto**.

```
# Login to the server with ssh
ubuntu@first-instance:~$ ssh ubuntu@10.196.36.97

# Become root
ubuntu@first-instance:~$ sudo su # I decided to just use sudo instead

# Check current disks
ubuntu@first-instance:~$ sudo fdisk -l

Disk /dev/sdd: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sde: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdf: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sdg: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

# Create new primary partion on each disk as type Linux raid auto that spans
over the whole disk
ubuntu@first-instance:~$ mdadm –help # Check help menu for the mdadm RAID tool
ubuntu@first-instance:~$ echo 'type=fd' | sudo sfdisk /dev/sdd

Checking that no-one is using this disk right now ... OK

Disk /dev/sdd: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
```

```
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

>>> Created a new DOS disklabel with disk identifier 0x90d78681.
/dev/sdd1: Created a new partition 1 of type 'Linux raid autodetect' and of
size 1023 MiB.
/dev/sdd2: Done.

New situation:
Disklabel type: dos
Disk identifier: 0x90d78681

Device     Boot Start     End Sectors  Size Id Type
/dev/sdd1        2048 2097151 2095104 1023M fd Linux raid autodetect

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

# Do the same for the other disks
ubuntu@first-instance:~$ sudo sfdisk /dev/sde
# tried without piping type=fd to see what happens
enter type=fd
printing results

ubuntu@first-instance:~$ echo 'type=fd' | sudo sfdisk /dev/sdf
ubuntu@first-instance:~$ echo 'type=fd' | sudo sfdisk /dev/sdg

# Check the disks after RAID partitioning
ubuntu@first-instance:~$ sudo fdisk -l

Disk /dev/sdd: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x90d78681

Device     Boot Start     End Sectors  Size Id Type
/dev/sdd1        2048 2097151 2095104 1023M fd Linux raid autodetect


Disk /dev/sde: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xeb724f45

Device     Boot Start     End Sectors  Size Id Type
/dev/sde1        2048 2097151 2095104 1023M fd Linux raid autodetect


Disk /dev/sdf: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x5a85a2fb

Device     Boot Start     End Sectors  Size Id Type
/dev/sdf1        2048 2097151 2095104 1023M fd Linux raid autodetect


Disk /dev/sdg: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xac610995

Device     Boot Start     End Sectors  Size Id Type
/dev/sdg1        2048 2097151 2095104 1023M fd Linux raid autodetect
```

Step 4: Install the necessary packages

- The following packages are needed for **RAID** and **LVM**. If they are not already installed on the system, install them as follows:

```
sudo apt -y install mdadm lvm2
```

- Reboot the system to make sure that all changes take effect

```
# Install necessary packages
ubuntu@first-instance:~$ sudo apt -y install mdadm lvm2
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded. # Already
installed
```

Step 5: Create a RAID Volume

- Login to the machine
- Become root with the **sudo** command
- Create a **RAID 5** volume called **/dev/md0** utilizing the new disks with the possibility of having **1 Hot-spare** disk

```
# Find out how mdadm works
ubuntu@first-instance:~$ mdadm --help
ubuntu@first-instance:~$ mdadm --help-options
ubuntu@first-instance:~$ mdadm --assemble --help

# Create RAID5 with 1 hot-spare disk
ubuntu@first-instance:~$ sudo mdadm --create -f /dev/md0 --level=5 –raid-
devices=3 /dev/sd[d-f]1
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.

# Check info of newly created RAID device
ubuntu@first-instance:~$ sudo mdadm -D /dev/md0

/dev/md0:
           Version : 1.2
     Creation Time : Wed Feb 21 11:16:19 2024
        Raid Level : raid5
        Array Size : 2091008 (2042.00 MiB 2141.19 MB)
     Used Dev Size : 1045504 (1021.00 MiB 1070.60 MB)
      Raid Devices : 3
     Total Devices : 3
       Persistence : Superblock is persistent

       Update Time : Wed Feb 21 11:16:26 2024
             State : clean
    Active Devices : 3
   Working Devices : 3
    Failed Devices : 0
     Spare Devices : 0

            Layout : left-symmetric
        Chunk Size : 512K

Consistency Policy : resync

              Name : first-instance:0  (local to host first-instance)
              UUID : bd813d56:22f72be4:9e5da3fe:8f351c59
            Events : 17

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       65        1      active sync   /dev/sde1
       2       8       81        2      active sync   /dev/sdf1

# I fucked up and it did not create any spare devices. So I need to remove the
RAID and try again
ubuntu@first-instance:~$ sudo umount /dev/md0
ubuntu@first-instance:~$ sudo mdadm --stop /dev/md0
ubuntu@first-instance:~$ sudo nvim /etc/mdadm/mdadm.conf
# Check if they are mentioned in config. They were not
```

```
ubuntu@first-instance:~$ sudo mdadm --remove /dev/md0
ubuntu@first-instance:~$ sudo mdadm --zero-superblock /dev/sd[d-g]1

# Create RAID5 with 1 hot-spare disk
ubuntu@first-instance:~$ sudo mdadm --create -f /dev/md0 --level=5 --raid-
devices=4 /dev/sd[d-g]1
# according to documentation, mdadm will make one disk a hot-spare when using
level 5.
ubuntu@first-instance:~$ sudo mdadm -D /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Feb 21 11:36:57 2024
        Raid Level : raid5
        Array Size : 3136512 (2.99 GiB 3.21 GB)
     Used Dev Size : 1045504 (1021.00 MiB 1070.60 MB)
      Raid Devices : 4
     Total Devices : 4
       Persistence : Superblock is persistent

       Update Time : Wed Feb 21 11:37:04 2024
             State : clean
    Active Devices : 4
   Working Devices : 4
    Failed Devices : 0
     Spare Devices : 0

            Layout : left-symmetric
        Chunk Size : 512K

Consistency Policy : resync

              Name : first-instance:0  (local to host first-instance)
              UUID : a93a8075:f8e39aad:5503ac15:945be703
            Events : 17

    Number   Major   Minor   RaidDevice State
       0       8       49        0      active sync   /dev/sdd1
       1       8       65        1      active sync   /dev/sde1
       2       8       81        2      active sync   /dev/sdf1
       3       8       97        3      active sync   /dev/sdg1

ubuntu@first-instance:~$ cat /proc/mdstat
Personalities : [linear] [multipath] [raid0] [raid1] [raid6] [raid5] [raid4]
[raid10]
md0 : active raid5 sdg1[3] sdf1[2] sde1[1] sdd1[0]
      3136512 blocks super 1.2 level 5, 512k chunk, algorithm 2 [4/4] [UUUU]

unused devices: <none>

# Add the RAID configuration to file
ubuntu@first-instance:~$ sudo bash -c 'mdadm --detail --scan | grep md0 >>
/etc/mdadm/mdadm.conf'
ubuntu@first-instance:~$ cat /etc/mdadm/mdadm.conf
# mdadm.conf
#
# !NB! Run update-initramfs -u after updating this file.
# !NB! This will ensure that initramfs has an uptodate copy.
#
# Please refer to mdadm.conf(5) for information about this file.
#
```

```
# by default (built-in), scan all partitions (/proc/partitions) and all
# containers for MD superblocks. alternatively, specify devices to scan, using
# wildcards if desired.
#DEVICE partitions containers

# automatically tag new arrays as belonging to the local system
HOMEHOST <system>

# instruct the monitoring daemon where to send mail alerts
MAILADDR root

# definitions of existing MD arrays

# This configuration was auto-generated on Mon, 28 Aug 2023 22:12:33 +0000 by
mkconf
ARRAY /dev/md0 metadata=1.2 name=first-instance:0
UUID=a93a8075:f8e39aad:5503ac15:945be703

# Update initial ram filesystem to assist kernel when os is booting
ubuntu@first-instance:~$ sudo update-initramfs -u
update-initramfs: Generating /boot/initrd.img-5.15.0-94-generic
```

Step 6: Creating Logical Volumes

You can of course use the newly created **RAID** disk **/dev/md0** by putting a filesystem on top of it, but in the following tasks we will utilize the capabilities of **Logical Volume Management** instead.

- Create a **Physical** Volume on **/dev/md0**:
  - Check that the Physical Volume is created
- Then create a **Volume Group** called **vg1** on **/dev/md0**:
  - Check that the Volume Group **vg1** is created
- Create two **Logical Volumes** (named **adm1**, and **adm2**, of size **200 MB**) on the volume group **vg1**
  - Check that both logical volumes are created

```
# Partition md0 to prepare for LVM
ubuntu@first-instance:~$ echo 'type=8e' | sudo sfdisk /dev/md0

Checking that no-one is using this disk right now ... OK

Disk /dev/md0: 2.99 GiB, 3211788288 bytes, 6273024 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 524288 bytes / 1572864 bytes

>>> Created a new DOS disklabel with disk identifier 0x737b2d06.
/dev/md0p1: Created a new partition 1 of type 'Linux LVM' and of size 3 GiB.
/dev/md0p2: Done.

New situation:
Disklabel type: dos
Disk identifier: 0x737b2d06

Device     Boot Start     End Sectors Size Id Type
/dev/md0p1       3072 6273023 6269952   3G 8e Linux LVM

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

# Check that md0 is partitioned
ubuntu@first-instance:~$ sudo fdisk -l

Disk /dev/md0: 2.99 GiB, 3211788288 bytes, 6273024 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 524288 bytes / 1572864 bytes
Disklabel type: dos
Disk identifier: 0x737b2d06

Device     Boot Start     End Sectors Size Id Type
/dev/md0p1       3072 6273023 6269952   3G 8e Linux LVM

# Create a physical volume on /dev/md0
ubuntu@first-instance:~$ sudo pvcreate /dev/md0p1

Physical volume "/dev/md0p1" successfully created.
```

```
# Check that the physical volume is created
ubuntu@first-instance:~$ sudo pvdisplay -C
  PV         VG Fmt  Attr PSize  PFree
  /dev/md0p1     lvm2 ---  <2.99g <2.99g

# Create a volume group, vg1, on /dev/md0
ubuntu@first-instance:~$ sudo vgcreate vg1 /dev/md0p1
 Volume group "vg1" successfully created

# Check that vg1 is created
ubuntu@first-instance:~$ sudo vgdisplay -C

VG  #PV #LV #SN Attr   VSize  VFree
  vg1   1   0   0 wz--n- <2.99g <2.99g

# Create two logical volumes, adm1 and adm2 with size 200MB, on volume group
vg1
ubuntu@first-instance:~$ sudo lvcreate -L 200MB -n adm1 vg1
  Logical volume "adm1" created.

ubuntu@first-instance:~$ sudo lvcreate -L 200MB -n adm2 vg1
 Logical volume "adm2" created.

# Check information about physical volume, volume group and logical volume
ubuntu@first-instance:~$ sudo pvdisplay
 --- Physical volume ---
  PV Name               /dev/md0p1
  VG Name               vg1
  PV Size               <2.99 GiB / not usable 0
  Allocatable           yes
  PE Size               4.00 MiB
  Total PE              765
  Free PE               665
  Allocated PE          100
  PV UUID               wXIh1n-ntqf-wUYu-mCZY-YpNb-ig0P-GsY7nD

ubuntu@first-instance:~$ sudo vgdisplay
--- Volume group ---
  VG Name               vg1
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  3
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               0
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <2.99 GiB
  PE Size               4.00 MiB
  Total PE              765
  Alloc PE / Size       100 / 400.00 MiB
  Free  PE / Size       665 / <2.60 GiB
  VG UUID               ag7qWj-bZ3j-91Wj-MudA-TADz-tHip-Zujys9
```

```
ubuntu@first-instance:~$ sudo lvdisplay
--- Logical volume ---
  LV Path                /dev/vg1/adm1
  LV Name                adm1
  VG Name                vg1
  LV UUID                r6LFd8-fs65-B3DX-dG1h-TaX5-lMNi-sZVCes
  LV Write Access        read/write
  LV Creation host, time first-instance, 2024-02-21 12:15:19 +0000
  LV Status              available
  # open                 0
  LV Size                200.00 MiB
  Current LE             50
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     6144
  Block device           253:0

  --- Logical volume ---
  LV Path                /dev/vg1/adm2
  LV Name                adm2
  VG Name                vg1
  LV UUID                UaOUXB-aCd1-jhsw-b9CI-ugSj-1L5N-ab298y
  LV Write Access        read/write
  LV Creation host, time first-instance, 2024-02-21 12:15:26 +0000
  LV Status              available
  # open                 0
  LV Size                200.00 MiB
  Current LE             50
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     6144
  Block device           253:1
```

Step 7: Creating filesystem on the Logical Volumes

- Create a filesystem of type **ext4** on the 2 logical volumes that you created.
- Create 2 directories named **adm1**, and **adm2** under the **/mnt** directory
- Mount the newly created **ext4** filesystems on **/mnt/adm1** and **/mnt/adm2** respectively.
- Issue the command **mount** to see if the filesystems are mounted on the appropriately:
- To mount the filesystems permanently upon startup/reboot, edit the file **/etc/fstab** and add the appropriately settings:
- Reboot the machine to see if everything is working fine and accordingly:
- Logical volumes are now ready to be used.
- Create a directory called **Dir-1** under **/mnt/adm1** and a directory called **Dir-2** under **/mnt/adm2**.
- Create a file called **file-1.txtHeading 1** under **Dir-1** a nd a file called **file-2.txt** under **Dir-2**.
- Put some content into the files **file-1.txt** and **file-2.txt**.

```
# Create filesystem type ext4 on the two logical volumes
ubuntu@first-instance:~$ sudo mkfs -t ext4 /dev/vg1/adm1

mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 51200 4k blocks and 51200 inodes
Filesystem UUID: fef2f157-663a-4487-821a-886f3be1dd1c
Superblock backups stored on blocks:
        32768

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

ubuntu@first-instance:~$ sudo mkfs -t ext4 /dev/vg1/adm2

mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 51200 4k blocks and 51200 inodes
Filesystem UUID: 1192cc5c-d1a0-416e-a51d-45a8fe8ac865
Superblock backups stored on blocks:
        32768

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

# check if filesystems was created
ubuntu@first-instance:~$ sudo blkid | grep adm

/dev/mapper/vg1-adm2: UUID="1192cc5c-d1a0-416e-a51d-45a8fe8ac865"
BLOCK_SIZE="4096" TYPE="ext4"
/dev/mapper/vg1-adm1: UUID="fef2f157-663a-4487-821a-886f3be1dd1c"
BLOCK_SIZE="4096" TYPE="ext4"

# Perform file system checks
ubuntu@first-instance:~$ sudo fsck /dev/vg1/adm1

fsck from util-linux 2.37.2
```

```
e2fsck 1.46.5 (30-Dec-2021)
/dev/mapper/vg1-adm1: clean, 11/51200 files, 7358/51200 blocks

ubuntu@first-instance:~$ sudo fsck /dev/vg1/adm2

fsck from util-linux 2.37.2
e2fsck 1.46.5 (30-Dec-2021)
/dev/mapper/vg1-adm2: clean, 11/51200 files, 7358/51200 blocks

# Create two directories named adm1 and adm2 under /mnt
ubuntu@first-instance:~$ sudo mkdir /mnt/adm1
ubuntu@first-instance:~$ sudo mkdir /mnt/adm2

# Mount the ext4 filesystems on /mnt/adm1 and /mnt/adm2
ubuntu@first-instance:~$ sudo mount -t ext4 /dev/vg1/adm1 /mnt/adm1
ubuntu@first-instance:~$ sudo mount -t ext4 /dev/vg1/adm2 /mnt/adm2

# use the mount command to see that the filesystems are mounted
ubuntu@first-instance:~$ mount
/dev/mapper/vg1-adm1 on /mnt/adm1 type ext4 (rw,relatime,stripe=384)
/dev/mapper/vg1-adm2 on /mnt/adm2 type ext4 (rw,relatime,stripe=384)

# Mount the systems permanently with /etc/fstab
ubuntu@first-instance:~$ blkid # get UUID for mounting
/dev/mapper/vg1-adm1: UUID="fef2f157-663a-4487-821a-886f3be1dd1c"
/dev/mapper/vg1-adm2: UUID="1192cc5c-d1a0-416e-a51d-45a8fe8ac865"

ubuntu@first-instance:~$ sudo nvim /etc/fstab

UUID='fef2f157-663a-4487-821a-886f3be1dd1c' /mnt/adm1      ext4
rw,relatime,nofail,auto 0 0
UUID='1192cc5c-d1a0-416e-a51d-45a8fe8ac865' /mnt/adm2      ext4
rw,relatime,nofail,auto 0 0

# Reboot to check if everything works
ubuntu@first-instance:~$ sudo reboot
ubuntu@first-instance:~$ ls /mnt # directories are there
ubuntu@first-instance:~$ mount | grep adm

/dev/mapper/vg1-adm1 on /mnt/adm1 type ext4 (rw,relatime,stripe=384)
/dev/mapper/vg1-adm2 on /mnt/adm2 type ext4 (rw,relatime,stripe=384)

# Create Dir-1 under /mnt/adm1 and Dir-2 under /mnt/adm2
ubuntu@first-instance:~$ sudo mkdir -p /mnt/adm1/Dir-1
ubuntu@first-instance:~$ tree /mnt
/mnt
├── adm1
│   ├── Dir-1
│   └── lost+found  [error opening dir]
└── adm2
    ├── Dir-2
    └── lost+found  [error opening dir]

# Create file-1.txt under Dir-1 and file-2.txt under Dir-2 and add content
ubuntu@first-instance:~$ cd /mnt/adm2/Dir-2
ubuntu@first-instance:~$ sudo touch file-2.txt
ubuntu@first-instance:~$ sudo chmod 777 file-2.txt
ubuntu@first-instance:~$ echo "hei sysadmin du er sexy" > file-2.txt
ubuntu@first-instance:~$ cd ../../adm1/Dir-1
```

```
ubuntu@first-instance:~$ sudo touch file-1.txt
ubuntu@first-instance:~$ echo «yeah boi yeah» > file-2.txt
```

Step 8: Simulating a failed disk

- Simulate a failed disk by turning off the Virtual Machine and removing one of the disks from the SATA/SCSI controller
    - You can also simulate a disk failure with the **mdadm -f** command. Whichever method you choose, the result will be the same
- Start the machine, login and observe the content of the file **/proc/mdstat**
    - Pay attention to the number of disks that are being constructed by the RAID array
- Read some data from the logical volumes.
    - Is the data still available?
- Turn off the machine again and reattach the disk you removed back to the SATA/SCSI controller
    - If you previously used the **mdadm -f** method to simulate a failure, you do not need to turn off the machine but rather use **mdadm** command to add the disk to the RAID array again
- Add the disk back to **RAID** volume **/dev/md0**
- Observe the content of the file **/proc/mdstat**
    - Pay attention to the number of disks that are being constructed by the **RAID** array

```
# Use mdadm to simulate disk failure on one disk
ubuntu@first-instance:~$ sudo mdadm --manage --set-faulty /dev/md0 /dev/sde1
mdadm: set /dev/sde1 faulty in /dev/md0

# Check /proc/mdstat and mdadm detail
ubuntu@first-instance:~$ cat /proc/mdstat
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1]
[raid10]
md0 : active raid5 sdd1[1] sdb1[3] sde1[0](F) sdc1[2]
      3136512 blocks super 1.2 level 5, 512k chunk, algorithm 2 [4/3] [_UUU]

unused devices: <none>

ubuntu@first-instance:~$ sudo mdadm --detail /dev/md0

/dev/md0:
           Version : 1.2
     Creation Time : Wed Feb 21 11:36:57 2024
        Raid Level : raid5
        Array Size : 3136512 (2.99 GiB 3.21 GB)
     Used Dev Size : 1045504 (1021.00 MiB 1070.60 MB)
      Raid Devices : 4
     Total Devices : 4
       Persistence : Superblock is persistent

       Update Time : Wed Feb 21 12:46:34 2024
             State : clean, degraded
    Active Devices : 3
   Working Devices : 3
    Failed Devices : 1
     Spare Devices : 0

            Layout : left-symmetric
        Chunk Size : 512K
```

```
Consistency Policy : resync

              Name : first-instance:0  (local to host first-instance)
              UUID : a93a8075:f8e39aad:5503ac15:945be703
            Events : 21

    Number   Major   Minor   RaidDevice State
       -       0        0        0      removed
       1       8       49        1      active sync   /dev/sdd1
       2       8       33        2      active sync   /dev/sdc1
       3       8       17        3      active sync   /dev/sdb1

       0       8       65        -      faulty   /dev/sde1


# Read data from the logical volumes
ubuntu@first-instance:~$ cat /mnt/adm1/Dir-1/file-1.txt
yeah boi yeah

ubuntu@first-instance:~$ cat /mnt/adm2/Dir-2/file-2.txt
hei sysadmin du er sexy

# Use mdadm to add disk back to the RAID array
ubuntu@first-instance:~$ sudo mdadm /dev/md0 -r /dev/sde1
ubuntu@first-instance:~$ sudo reboot
ubuntu@first-instance:~$ sudo mdadm /dev/md0 -a /dev/sde1
mdadm: added /dev/sde1

# Check /proc/mdstat and mdadm detail
ubuntu@first-instance:~$ sudo mdadm --detail /dev/md0
/dev/md0:
           Version : 1.2
     Creation Time : Wed Feb 21 11:36:57 2024
        Raid Level : raid5
        Array Size : 3136512 (2.99 GiB 3.21 GB)
     Used Dev Size : 1045504 (1021.00 MiB 1070.60 MB)
      Raid Devices : 4
     Total Devices : 4
       Persistence : Superblock is persistent

       Update Time : Wed Feb 21 12:57:16 2024
             State : clean
    Active Devices : 4
   Working Devices : 4
    Failed Devices : 0
     Spare Devices : 0

            Layout : left-symmetric
        Chunk Size : 512K

Consistency Policy : resync

              Name : first-instance:0  (local to host first-instance)
              UUID : a93a8075:f8e39aad:5503ac15:945be703
            Events : 48

    Number   Major   Minor   RaidDevice State
       4       8       65        0      active sync   /dev/sde1
       1       8       49        1      active sync   /dev/sdd1
```

```
       2       8       33        2      active sync   /dev/sdc1
       3       8       17        3      active sync   /dev/sdb1

ubuntu@first-instance:~$ cat /proc/mdstat
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1]
[raid10]
md0 : active raid5 sde1[4] sdd1[1] sdb1[3] sdc1[2]
      3136512 blocks super 1.2 level 5, 512k chunk, algorithm 2 [4/4] [UUUU]

unused devices: <none
```

Step 9: Resizing LVM and the filesystem

- Assume that **/mnt/adm1** has grown more than we predicted. Expand this logical volume with more space (E.g. **50 MB** more)

```
# check the logical volume adm1
ubuntu@first-instance:~$ sudo fdisk -l

Disk /dev/mapper/vg1-adm1: 200 MiB, 209715200 bytes, 409600 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 524288 bytes / 1572864 bytes

ubuntu@first-instance:~$ sudo lvdisplay
 --- Logical volume ---
  LV Path                /dev/vg1/adm1
  LV Name                adm1
  VG Name                vg1
  LV UUID                r6LFd8-fs65-B3DX-dG1h-TaX5-lMNi-sZVCes
  LV Write Access        read/write
  LV Creation host, time first-instance, 2024-02-21 12:15:19 +0000
  LV Status              available
  # open                 1
  LV Size                200.00 MiB
  Current LE             50
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     6144
  Block device           253:

# Add 50MB of space to the logical volume
ubuntu@first-instance:~$ sudo lvextend --size +50MB -r /dev/vg1/adm1

Rounding size to boundary between physical extents: 52.00 MiB.
  Size of logical volume vg1/adm1 changed from 200.00 MiB (50 extents) to
252.00 MiB (63 extents).
  Logical volume vg1/adm1 successfully resized.
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/mapper/vg1-adm1 is mounted on /mnt/adm1; on-line resizing
required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/mapper/vg1-adm1 is now 64512 (4k) blocks long.

# Check adm1
ubuntu@first-instance:~$ sudo lvdisplay
--- Logical volume ---
  LV Path                /dev/vg1/adm1
  LV Name                adm1
  VG Name                vg1
  LV UUID                r6LFd8-fs65-B3DX-dG1h-TaX5-lMNi-sZVCes
  LV Write Access        read/write
  LV Creation host, time first-instance, 2024-02-21 12:15:19 +0000
  LV Status              available
  # open                 1
  LV Size                252.00 MiB
  Current LE             63
  Segments               2
```

```
  Allocation              inherit
  Read ahead sectors      auto
  - currently set to      6144
  Block device            253:0

ubuntu@first-instance:~$ sudo fdisk -l

Disk /dev/mapper/vg1-adm1: 252 MiB, 264241152 bytes, 516096 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 524288 bytes / 1572864 bytes
```

Assignment-4: Software Installation (dpkg, apt and from source)

The goal of this week's assignment is to:

- understand the package management tool used by Ubuntu.
- learn to search, install and remove software with the package management tool.
- learn to install software from source.
- install software with the **checkinstall** utility.

As the newly recruited system administrator you receive the following mail from the senior system administrator:

Hi Novice Sysadmin,

Great work on getting **RAID** and **LVM** on the newly purchased server. We would like you to get acquainted with how to install and manage software on Ubuntu. Sometimes, we also need to install software from source so it is crucial that you get accustomed to them by doing the steps on the following pages.

We also advise you to explore and practice even more on how to manage software with **dpkg**, **apt** and from source as having a very good grasp on these as a system administrator will ease the process of managing software on our servers.

A good resource to start with is:

- Chapter 20 of Linux System Administration by Paul Cobbaut
    - http://linux-training.be/sysadmin/ch20.html
- Chapter 12 of Linux System Administration I by Tuxcademy
    - https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
- Package Management by Ubuntu's website
    - https://ubuntu.com/server/docs/package-management
    - https://help.ubuntu.com/community/InstallingSoftware
    - https://help.ubuntu.com/community/CompilingSoftware

Best regards
Senior Sysadmin!

Step 1: Installing software without dependencies with dpkg

In this section, you will download .deb packages from Ubuntu website and install it with the built-in package management tool dpkg

1. First, list all the packages installed on your system (Virtual machine) with **dpkg -l**
2. Let's first download a **.deb** package that doesn't have any other dependencies: Download the **cowsay** package from: http://archive.ubuntu.com/ubuntu/pool/universe/c/cowsay

   wget http://archive.ubuntu.com/ubuntu/pool/universe/c/cowsay/cowsay_3.03+dfsg2-4_all.deb

1. Install the package with the **dpkg** command:

   sudo dpkg -i cowsay_3.03+dfsg2-4_all.deb

2. Verify that **cowsay** is installed on the system:

   dpkg -l| grep cowsay

3. Now run the **cowsay** command ($ **cowsay "Hello World!"**):

```
# list packages
ubuntu@first-instance:~$ dpkg -l

Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                              Version
Architecture Description
+++-==============================-=====================================-
============-========================>
ii  adduser                           3.118ubuntu5                          all
add and remove users and >
ii  apparmor                          3.0.4-2ubuntu2.3
amd64        user-space parser utility>
ii  apport                            2.20.11-0ubuntu82.5                   all
automatically generate cr>
ii  apport-symptoms                   0.24                                  all
symptom scripts for apport

...

# Fetch cowsay package from ubuntu repository
ubuntu@first-instance:~$ wget
http://archive.ubuntu.com/ubuntu/pool/universe/c/cowsay/cowsay_3.03+dfsg2-
4_all.deb

# install package with dpkg
ubuntu@first-instance:~$ sudo dpkg -i cowsay_3.03+dfsg2-4_all.deb
```

```
# Check if package is installed
ubuntu@first-instance:~$ dpkg -l| grep cowsay

# Run cowsay
ubuntu@first-instance:~$ cowsay "hello bitch"
_____________

< hello bitch >
 -------------
        \   ^__^
         \  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
```

Step 2: Installing software with dependencies with dpkg

1. Now download a **.deb** package that has dependencies: Download the text editor **jed** package from: https://packages.ubuntu.com/jammy/jed

The page will give you the possibility to download **jed** from a mirror site. Choose the following:

Openstack Cloud
wget https://no.archive.ubuntu.com/ubuntu/pool/universe/j/jed/jed_0.99.20~pre.158%2Bdfsg-1_amd64.deb

1. Explore the package information and find out what the dependencies are:

   # Openstack Cloud
   sudo dpkg --info jed_0.99.20~pre.158+dfsg-1_amd64.deb

2. The output shows that jed depends on among other things jed-common: (NB!, if you try to install jed before the dependencies are installed, you will get an error!)
3. Install **jed-common** without any recommendations through the package management system:

   sudo apt-get --no-install-recommends install jed-common

4. Now, install the package jed that you downloaded earlier with the dpkg command:

   # Openstack Cloud
   sudo dpkg -i jed_0.99.20~pre.158+dfsg-1_amd64.deb

5. Verify that **jed** is installed on the system:

   dpkg -l jed

6. Now run the **jed** command (**To quit jed, hold ctrl x c**):

Congratulation! You have managed to download and install **.deb** packages with the built-in package management tool **dpkg.**

```
# Download jed
ubuntu@first-instance:~$ wget
https://no.archive.ubuntu.com/ubuntu/pool/universe/j/jed/jed_0.99.20~pre.158%2
Bdfsg-1_amd64.deb

# Explore package information and dependencies
ubuntu@first-instance:~$ sudo dpkg --info jed_0.99.20~pre.158+dfsg-
1_amd64.deb4

new Debian package, version 2.0.
 size 167838 bytes: control archive=1791 bytes.
    1420 bytes,    33 lines      control
     588 bytes,     9 lines      md5sums
     816 bytes,    37 lines   *  postinst             #!/bin/sh
     275 bytes,    13 lines   *  prerm                #!/bin/sh
```

```
 Package: jed
 Version: 1:0.99.20~pre.158+dfsg-1
 Architecture: amd64
 Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
 Original-Maintainer: Wookey <wookey@debian.org>
 Installed-Size: 444
 Depends: libc6 (>= 2.34), libgpm2 (>= 1.20.7), libslang2 (>= 2.2.4), jed-
common (= 1:0.99.20~pre.158+dfsg-1)
 Suggests: gpm
 Provides: editor, info-browser, mail-reader
 Section: editors
 Priority: optional
 Homepage: https://www.jedsoft.org/jed/
 Description: editor for programmers (textmode version)
  Jed is a small, fast and powerful text editor, yet starts faster than
  bash.
  .
  Completely customizable with prepared emulation modes for Emacs, CUA
  (similar to KDE/Gnome/OpenOffice), Borland-IDE, Brief, and EDT. Extensible
  in the S-Lang scripting language (with a syntax resembling C).
  .
  Editing functions: folding support, rectangular cut/paste, regular
  expression search/replace, incremental searches, search/replace across
  multiple files, multiple windows, multiple buffers, ...
  .
  Tools: directory editor (dired), info (browse GNU info files), mail,
  rmail, ispell, shell mode, ...
  .
  Special modes (syntax highlight, indention, compile, ...) for
  Basic, C, C++, DCL, FORTRAN, IDL, Java, nroff, Pascal, Perl, PHP,
  PostScript, Python, sh.  Modes for markup languages include HTML and
  (La)TeX (with AUC-TeX style editing and BibTeX)
  .
  Additional tools and modes can be found in the jed-extra package.

# install jed-common with apt and without recommendations
ubuntu@first-instance:~$ sudo apt-get --no-install-recommends install jed-
common

# Install the downloaded package
ubuntu@first-instance:~$ sudo dpkg -i jed_0.99.20~pre.158+dfsg-1_amd64.deb

# Check if it was installed
ubuntu@first-instance:~$ dpkg -l jed

Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name           Version                  Architecture Description
+++-==============-========================-============-
=========================================
ii  jed            1:0.99.20~pre.158+dfsg-1 amd64        editor for programmers
(textmode version)

# Try jed
ubuntu@first-instance:~$ jed

ctrl+x+c # to quit
```

Step 3: Exploring the apt package management tool used by Ubuntu

In this section, you will search, install and remove software with the built-in package management system **apt**

1. The first thing you should do is update the system so that you get the latest version of the software

   sudo apt update -y; sudo apt-get clean -y

2. We would like you to install the Nginx web server with the package management tool **apt**. Search for nginx:

   apt search nginx

3. Then install Nginx

   sudo apt install nginx -y

   NB! If your nginx cannot be started and throws some errors regarding nginx: [emerg] socket() [::]:80 failed (97: Address family not supported by protocol). You have to edit /etc/nginx/sites-enabled/default and comment out this line listen [::]:80 default_server; then sudo systemctl start nginx

4. Test if the Nginx server is running and serving pages by executing the curl command. **Curl** is a handy utility that helps as download resources from a web server:

   curl -Is localhost:80

5. You should get something similar to the following output.

   HTTP/1.1 200 OK
   Server: nginx/1.18.0 (Ubuntu)
   Date: Mon, 27 Dec 2021 13:49:46 GMT
   Content-Type: text/html
   Content-Length: 612
   Last-Modified: Mon, 27 Dec 2021 13:49:25 GMT
   Connection: keep-alive
   ETag: "61c9c465-264"
   Accept-Ranges: bytes

6. From the output, we can see the version that gets installed from Ubuntu repository is **1.18.0**. You can also check the installed version with the command:

   dpkg -l |grep nginx

7. This is an old version compared to the latest version from Nginx website. We will in the next section install the latest version from source. For now remove nginx totally from the system:

   sudo apt -y remove --purge nginx nginx-common; sudo apt autoremove -y

8. Verify that Nginx is removed from the system:

   dpkg -l nginx

```
# Update system
ubuntu@first-instance:~$ sudo apt update -y; sudo apt-get clean -y

# Search for nginx in apt
ubuntu@first-instance:~$ apt search nginx

# Install nginx
ubuntu@first-instance:~$ sudo apt install nginx -y

# Check if nginx works with curl
ubuntu@first-instance:~$ curl -Is localhost:80

HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 21 Feb 2024 14:23:53 GMT
Content-Type: text/html
Content-Length: 612
Last-Modified: Wed, 21 Feb 2024 14:22:32 GMT
Connection: keep-alive
ETag: "65d60728-264"
Accept-Ranges: bytes

# check which version is installed
ubuntu@first-instance:~$ dpkg -l |grep nginx

ii  nginx                          1.18.0-6ubuntu14.4 amd64        small,
powerful, scalable web/proxy server


# Remove and purge nginx from the server
ubuntu@first-instance:~$ sudo apt -y remove --purge nginx nginx-common; sudo
apt autoremove -y

# Verify that it is removed
ubuntu@first-instance:~$ dpkg -l nginx

dpkg-query: no packages found matching nginx
```

Step 4: Installing software from source

In this section, you will install an Nginx HTTP server from source. Before software can be installed from source, Ubuntu needs some necessary packages for compiling software. This is not installed by default and you get them by executing the following:

sudo apt install build-essential

1. Download the latest stable Nginx HTTP server (as of **11.Feb.2024**, the latest version is **1.24.0**) from: http://nginx.org/en/download.html
   wget https://nginx.org/download/nginx-1.24.0.tar.gz

1. Unpack the archive, change to the source directory and read the INSTALL file.

2. Explore the packages you need to install (according to the INSTALL file) before you configure and compile Nginx?

a. You can also get detailed info from Nginx site:
http://nginx.org/en/docs/configure.html

4. Run the configure command with the help option. Make sure you're in the proper directory when you do this.
   ./configure --help

5. Once you are confident about what options you want to configure the software with and all the dependencies are met proceed to configure it:

   ./configure

   NB!
   Configure will complain about missing dependencies such as **PCRE** and **ZLIB**. Make sure that Nginx is configured with the required dependencies.
   Make also sure that Nginx is configured with **Openssl**

6. Compile Nginx with the make command.

   make

   NB! If make does not give you any errors, continue installing it.

7. Install Nginx
   sudo make install

8. Test if the Nginx HTTP server is running and serving pages by executing the command (if you don't get any output, it means the server is not running):
   curl -Is localhost:80

9. Start Nginx HTTP server by executing the command:
   sudo /usr/local/nginx/sbin/nginx

10. Test if the Nginx server is running and serving pages by executing the command:
    curl -Is localhost:80
    You should get some info about Nginx

```
# Install build-essential to be able to build packages from source
ubuntu@first-instance:~$ sudo apt install build-essential

# Install the latest nginx package from source
ubuntu@first-instance:~$ wget https://nginx.org/download/nginx-1.24.0.tar.gz

# Unpack the archive
ubuntu@first-instance:~$ tar -xvzf nginx-1.24.0.tar.gz

# change directory
ubuntu@first-instance:~$ cd nginx-1.24.0

# No install file, README says to read documentation on their site
ubuntu@first-instance:~$ ./configure --help

# Run configure with no options to do default
ubuntu@first-instance:~$ ./configure

# Gives error because of lacking PCRE library

# Run configure with necessary options
ubuntu@first-instance:~$ ./configure –without-http_rewrite_module

# missing zlib and gzip library

# Run again with more options
ubuntu@first-instance:~$ ./configure –without-http_rewrite_module –without-
http_gzip_module

# compile nginx with make
ubuntu@first-instance:~$ make

# install nginx
ubuntu@first-instance:~$ sudo make install

# Test nginx
ubuntu@first-instance:~$ curl -Is localhost:80 # no results

# Start nginx
ubuntu@first-instance:~$ sudo /usr/local/nginx/sbin/nginx

# Test again
ubuntu@first-instance:~$ curl -Is localhost:80

HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Wed, 21 Feb 2024 14:45:39 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Wed, 21 Feb 2024 14:44:34 GMT
Connection: keep-alive
ETag: "65d60c52-267"
Accept-Ranges: bytes
```

Step 5: Installing software with *checkinstall*

In this section, you will install an Nginx HTTP server with the **checkinstall** utility. The **checkinstall** utility will create and install **.deb** package managed by the built-in package manager **dpkg**.

1. You need to first Install the **checkinstall** utility

   sudo apt install checkinstall

2. Now stop the running Nginx server that you installed in the previous step:

   sudo killall -9 nginx

3. Repeat steps **4.5** and **4.6** in the previous Step 4 section:

○ You should basically **configure** nginx with the required dependencies and compile it with **make**

NB!
Do not execute make install this time as you should be using **checkinstall** instead

4. Instead of executing "make install" run **checkinstall** . Make sure that you are still in the same directory.

   sudo checkinstall

5. Accept all the defaults presented to you as this will install the **.deb** package created for your system. You may add your own package description if you like.
6. Verify that the **nginx** is installed on the system by executing

   dpkg -l |grep nginx

7. Test if Nginx HTTP server is running and serving pages by executing the command (if you don't get any output, it means the server is not running):

   curl -Is localhost:80

8. Start Nginx HTTP server by executing the command:

   sudo /usr/local/nginx/sbin/nginx

9. Test if the Nginx server is running and serving pages by executing the command:

   curl -Is localhost:80

You should get some info about nginx:

Congratulation, you managed to install software with checkinstall!!!

```
# Install checkinstall
ubuntu@first-instance:~$ sudo apt install checkinstall

# Stop running nginx

ubuntu@first-instance:~$ sudo killall -9 nginx

# configure nginx again

ubuntu@first-instance:~$ ./configure --without-http_rewrite_module –without-
http_gzip_module

# compile nginx again

ubuntu@first-instance:~$ make

# run checkinstall

ubuntu@first-instance:~$ sudo checkinstall # accept all defaults

# Verify install

ubuntu@first-instance:~$ dpkg -l |grep nginx

ii  nginx                          1.24.0-1 amd64        nginx


# Test if nginx server is running

ubuntu@first-instance:~$ curl -Is localhost:80 # not running

# Start nginx server

ubuntu@first-instance:~$ sudo /usr/local/nginx/sbin/nginx

# Test nginx server again

ubuntu@first-instance:~$ curl -Is localhost:80

HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Wed, 21 Feb 2024 14:50:56 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Wed, 21 Feb 2024 14:44:34 GMT
Connection: keep-alive
ETag: "65d60c52-267"
Accept-Ranges: bytes
```

Assignment-5: User management (Local Accounts)

The goal of this week's assignment is to:

- Understanding the user and group concepts of Linux
- Knowing how user and group information is stored on Linux
- Being able to use the user and group administration commands

---

As the newly recruited system administrator you receive the following mail from the senior system administrator:

Hi Novice Sysadmin,

the following employees have started at the company recently and are in need of accounts urgently. See the below table for more details.

First name	Last name	Username	Email	Tel	Group	Shell
James T.	Kirk	kirk	kirk@3610.no	11111111	captains	/usr/bin/fish
Bart	Simpson	bart	bart@3610.no	22222222	media	/bin/bash
Luke	Skywalker	luke	luke@3610.no	33333333	rebels	/bin/bash

We also advise you to explore and practice even more on how to manage accounts in Linux and get a very good grasp on these as a system administrator will ease the process of managing users.

A good resource to start with is:

- **Chapter 2** of Linux System Administration I by Tuxcademy
  https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
- **Chapter 28, 29, 30 and 31** of Linux System Administration by Paul Cobbaut
  http://linux-training.be/funhtml/pt08.html

Best regards
Senior Sysadmin!

Step 1: Get information about the users on the system

Finding information about users on the system is a necessary task for system administrators.

1. Find the **UID**, **GID** and the **SHELL** for the following users
   a. root
   b. nobody
   c. sysadmin
   d. www-data

```
# Task a - root
# Find root in passwd – Find shell, UID and GID
ubuntu@first-instance:~$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash

# Use id to find UID
ubuntu@first-instance:~$ id -u root
0

# Use id to find all GID
ubuntu@first-instance:~$ id -G root
0

# Find group 0 in group
ubuntu@first-instance:~$ cat /etc/group
root:x:0:

# root has UID and GID 0 (root), and uses /bin/bash as shell

# Task b – nobody
# Find nobody in passwd
ubuntu@first-instance:~$ cat /etc/passwd | grep nobody
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin

# Find group name for 65534
ubuntu@first-instance:~$ cat /etc/group | grep 65534
nogroup:x:65534:

# nobody has UID and GID 65535 (nogroup) and uses /usr/sbin/nologin as shell
(replacement to not be able to login)

# Task c – sysadmin
ubuntu@first-instance:~$ cat /etc/passwd | grep sysadmin
ubuntu@first-instance:~$ id -u sysadmin
id: ‘sysadmin’: no such user

# No user sysadmin

# Task d – www-data
# Find www-data in passwd
ubuntu@first-instance:~$ cat /etc/passwd | grep www-data
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin

# Find group name in group
ubuntu@first-instance:~$ cat /etc/group | grep 33
www-data:x:33:
```

Result:

Username	UID	GID	Shell
root	0	0 (root)	/bin/bash
nobody	65534	65534 (nogroup)	/usr/sbin/nologin
sysadmin	-	-	-
www-root	33	33 (www-data)	/usr/sbin/nologin

2. Find out if there is a user named **postfix** on the system

```
ubuntu@first-instance:~$ cat /etc/passwd | grep postfix
ubuntu@first-instance:~$ id -u postfix
id: ‘postfix’: no such user
```

3. Find out if there is a user named **backup** on the system

```
ubuntu@first-instance:~$ cat /etc/passwd | grep backup
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
```

4. Find out if there is a group named **lpadmin** on the system

```
ubuntu@first-instance:~$ cat /etc/group | grep lpadmin
# no group
```

5. **Find out if there is a group named dave3610 on the system**

```
ubuntu@first-instance:~$ cat /etc/group | grep dave3610
# no group
```

Step 2: Create the groups

- The first thing you need to do is create the necessary groups as mentioned in the mail from the **Senior Sysadmin**.

```
# Check if the groups captains, media, rebels are present
ubuntu@first-instance:~$ cat /etc/group | grep captains
ubuntu@first-instance:~$ cat /etc/group | grep media
ubuntu@first-instance:~$ cat /etc/group | grep rebels

# None of the groups present. Create the groups with groupadd
ubuntu@first-instance:~$ sudo groupadd media
ubuntu@first-instance:~$ sudo groupadd rebels
ubuntu@first-instance:~$ sudo groupadd captains

# Check if the groups are added to /etc/group
ubuntu@first-instance:~$ cat /etc/group | grep captains
captains:x:1001:
ubuntu@first-instance:~$ cat /etc/group | grep media
media:x:1002:
ubuntu@first-instance:~$ cat /etc/group | grep rebels
rebels:x:1003:
```

Step 3: Create the users

Create the necessary users as mentioned in the mail from the **Senior Sysadmin**. In addition, the following requirements must be taken into account.

First name	Last name	Username	Email	Tel	Group	Shell
James T.	Kirk	kirk	kirk@3610.no	11111111	captains	/usr/bin/fish
Bart	Simpson	bart	bart@3610.no	22222222	media	/bin/bash
Luke	Skywalker	luke	luke@3610.no	33333333	rebels	/bin/bash

1. Generate random passwords for each user.
2. All the users home directories must be under **/home**.
3. Check if the desired shell is installed and available in the system. Install it if it is not installed.

```
# Install pwgen to generate random password and fish shell for kirk
ubuntu@first-instance:~$ sudo apt install pwgen
ubuntu@first-instance:~$ sudo apt install fish

# Add kirk with useradd (captains is GID 1001)
ubuntu@first-instance:~$ sudo useradd -g 1001 -m -c "James T. Kirk, kirk@3610,
11111111" -s /usr/bin/fish kirk
ubuntu@first-instance:~$ cat /etc/passwd | grep kirk
kirk:x:1001:1001:James T. Kirk:/home/kirk:/usr/bin/fish

# Add bart with useradd (media GID is 1002)
ubuntu@first-instance:~$ sudo useradd -g 1002 -m -c "Bart Simpson,
bart@3610.no, 22222222" -s /bin/bash bart
ubuntu@first-instance:~$ cat /etc/passwd | grep bart
bart:x:1002:1002:Bart Simpson:/home/bart:/bin/bash

# Add luke with useradd (rebels GID is 1003)
ubuntu@first-instance:~$ sudo useradd -g 1003 -m -c "Luke Skywalker,
luke@3610.no, 33333333" -s /bin/bash luke
ubuntu@first-instance:~$ cat /etc/passwd | grep luke
luke:x:1003:1003:Luke Skywalker:/home/luke:/bin/bash

# Fix random password for the users
ubuntu@first-instance:~$ pwgen -B -N 3 > pwd.txt
ubuntu@first-instance:~$ cat pwd.txt
JaiK9oe7
UhooPh9n
fai9Yiez
ubuntu@first-instance:~$ nvim pwd.txt # change each line to username:password
ubuntu@first-instance:~$ sudo chpasswd < pwd.txt

# I fucked up the gecos field when adding users, so I had to go back and use
`sudo usermod -c <name, mail, number>` for all users

# I could have generated all three by creating a text file and used `newusers`
```

Step 4: Creating users in a bulk

Creating users individually one at a time is time-consuming. A sysadmin will be required to create users in a bulk. Assume that a program/script gets the details from a database and generates a file containing details of the users. Your job is to do the following:

1. Here is users' information:
   user1:6001:grp1:This is user1:/home/user1:/bin/bash
   user2:6002:grp2:This is user2:/home/user2:/bin/dash
   user3:6003:grp3:This is user3:/home/user3:/usr/bin/fish
   user4:6004:grp4:This is user4:/home/user4:/bin/bash
   user5:6005:grp5: gThis is user5:/home/user5:/usr/bin/fish
   user6:6006:grp6:This is user6:/home/user6:/bin/bash
   user7:6007:grp7:This is user7:/home/user7:/bin/false
   user8:6008:grp8:This is user8:/home/user8:/bin/sh

2. Here are user credentials
   user1:veeg9eip
   user2:Veigh9ai
   user3:shaChup4
   user4:gi9aeNg4
   user5:Eeg3ua7r
   user6:ohneiTe3
   user7:phoi7ieC
   user8:To3hah7U

3. Write a small bash script that creates the users in a bulk.
   Make sure that the passwords are hashed with SHA512 algorithm.

```
# First create a text file for user information and one for user credentials
ubuntu@first-instance:~$ nvim userinfo.txt
ubuntu@first-instance:~$ nvim usercred.txt

# Create script for automaticly generating users
ubuntu@first-instance:~$ nvim usergen.sh

```
#! /bin/bash

# Script for adding users in bulk

# Create new file
> newusers.txt

# Declare array with added groups
declare -A added_groups

# While loop which uses IFS to set variables for each entrance between :
while IFS=: read -r username uid group gecos home shell
do

      # Check if group has been added, if not add it
      if [ -z "${added_groups[$group]}" ]
      then
            if ! groupadd "$group"
            then
                  echo "Group '$group' already exists. Group not added"
            else
                  added_groups[$group]=1 # Mark group as added
            fi
      fi

      # Extract the password from usercred.txt
      # Using cut, picking : as delimiter and specifing field 2
      password=$(grep "^$username:" usercred.txt | cut -d':' -f2)

      # Hash the passwords using openssl with SHA512 (option -6 is sha512)
      hashed_pw=$(openssl passwd -6 "$password")

      # Append each line entry with password added to a new file
      echo "$username:$hashed_pw:$uid:$group:$gecos:$home:$shell" >>
newusers.txt
      echo "Added $username to file"
done < userinfo.txt

echo "Added all users to file. Creating users in bulk with newusers"
# Add users in bulk with newusers
newusers newusers.txt
```

Change execution rights and run script with sudo
ubuntu@first-instance:~$ chmod 755 usergen.sh
ubuntu@first-instance:~$ sudo ./usergen.sh
Added user1 to file
Added user2 to file
Added user3 to file
Added user4 to file
Added user5 to file
Added user6 to file
Added user7 to file
```

```
Added user8 to file
Added all users to file. Creating users in bulk with newusers
ubuntu@first-instance:~$ cat /etc/passwd
[…]
kirk:x:1001:1001:James T. Kirk, kirk@3610.no, 11111111:/home/kirk:/usr/bin/fish
bart:x:1002:1002:Bart Simpson, bart@3610.no, 11111111:/home/bart:/bin/bash
luke:x:1003:1003:Luke Skywalker, luke@3510, 333333333:/home/luke:/bin/bash
user1:x:6001:1004:This is user1:/home/user1:/bin/bash
user2:x:6002:1005:This is user2:/home/user2:/bin/dash
user3:x:6003:1006:This is user3:/home/user3:/usr/bin/fish
user4:x:6004:1007:This is user4:/home/user4:/bin/bash
user5:x:6005:1008:This is user5:/home/user5:/usr/bin/fish
user6:x:6006:1009:This is user6:/home/user6:/bin/bash
user7:x:6007:1010:This is user7:/home/user7:/bin/false
user8:x:6008:1011:This is user8:/home/user8:/bin/sh
```

Step 5: Account management

Knowing how to lock, disable, modify and delete an account is an essential part of system administration:

1. Make **kirk** is also a member of the **rebels** group in addition to his **primary group**.

```
# Use usermod to add secondary group
ubuntu@first-instance:~$ sudo usermod -G rebels kirk

# Check which groups kirk is a part of
ubuntu@first-instance:~$ id kirk
uid=1001(kirk) gid=1001(captains) groups=1001(captains),1003(rebels)
```

2. Lock the account for the user **user1** created in the previous step

```
# Lock the user with usermod
ubuntu@first-instance:~$ sudo usermod -L user1
```

3. Set the account expiry date for user **bart** to 1st of Mars, 2024

```
# Use chage to set account expiry
ubuntu@first-instance:~$ sudo chage -E 2024-03-01 user1
```

4. Delete **user8** from the system completely

```
# Use userdel to delete user8 and its home directory
ubuntu@first-instance:~$ sudo userdel -r user8
```

Assignment-6: Centralized Account Management and Quotas.

The **DAVE3610** University wants to implement a centralized account management such that all the users created whether they are staff, students are managed centrally. The University policy is to only use open standards and avoid proprietary solutions. The IT department has been given a task to build a solution based on Open Source technologies with the following specifications. The system must implement:

- A scalable **LDAP** server:
  - **OpenLDAP** is the required **LDAP** server for the University.
  - Users should be able to login with the same **username** and **password** from any Linux machine.
  - This includes the **LDAP** server as well. In other words the **LDAP** server must be configured as an **LDAP** client.

Hi Junior Sysadmin,

The following employees have started at the company recently and are in need of **centralized accounts** urgently. It is essential that the users are created on the **LDAP** server. See the below table for more details.

First name	Last name	Desired username	Email	Department/ Primary Group	Desired shell
Ismail	Hassan	ismail	ismail.hassan@dave3610.com	engineering	bash
Amir	Maqbool	amir	amir.maqbool@dave3610.com	sysadmins	bash
Geir	Skjevling	geir	geir.skjevling@dave3610.com	marketing	bash

A useful resource for this assignment is:

- https://ubuntu.com/server/docs/service-ldap
- https://www.openldap.org/doc/admin24/

Best regards
Senior Sysadmin!.

Step 1: Provision instances on OpenStack Cloud

For this assignment, you will need 2 instances. One can be an LDAP server and another can be an LDAP client. **We have no control over IP addresses**. Please note them down for a later use.

first_instance (server) / ip: 10.196.36.97ubuntu@first-instance:~$

ldap_client (client) / ip: 10.196.36.113

Step 2: Setup the LDAP server and the LDAP client

- Setup an **OpenLDAP** server and make sure that it uses:
  - suffix: dc=dave3610,dc=com
  - rootdn: cn=admin,dc=dave3610,dc=com
  - rootpw: pagh3377 hashed with {md5}

```
# Install slapd and ldap-utils
ubuntu@first-instance:~$ sudo apt install slapd ldap-utils
Configuring slapd
Please enter the admin password for your LDAP directory again to verify that
you have typed it correctly.
Confirm password: ******

# Configure suffix (base DN) for our server
ubuntu@first-instance:~$ sudo dpkg-reconfigure slapd

# Choose dave3610.com as DNS domain name, dave3610 as org name, do not purge,
move old db

# Query the database to check that we have new suffix
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:// -b dc=dave3610,dc=com

dn: dc=dave3610,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dave3610
dc: dave3610

# Verify the result
ubuntu@first-instance:~$ ldapsearch -x -LLL -D "cn=admin,dc=dave3610,dc=com" -W
-b "dc=dave3610,dc=com" "(objectClass=*)"
dn: dc=dave3610,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dave3610
dc: dave3610

~~~~~~

# Turns out I fucked up the admin password by using my root password and not
pagh3377 as in the demo. So to fix this I did the slapd setup again with
correct password:
sudo dpkg-reconfigure slapd # Remove when purged – no, move old db – yes

# Change the passwd file to pagh3377 again
ubuntu@first-instance:~$ sudo sh -c 'echo -n 'pagh3377' >
/etc/ldapscripts/ldapscripts.passwd'

~~~~
# Add admin user and hash password
ubuntu@first-instance:~$ slappasswd -h {MD5} -s pagh3377
```

```
ubuntu@first-instance:~$ nvim admin.ldif
dn: cn=admin,dc=dave3610,dc=com
objectClass: simpleSecurityObject
objectClass: organizationalRole
cn: admin
userPassword: {md5}yoZlaWcO7H++3KLQBM5FUw==
description: LDAP administrator

# Add admin user
ubuntu@first-instance:~$ ldapadd -x -D "cn=admin,dc=dave3610,dc=com" -W -f
admin.ldif
Enter LDAP Password:
adding new entry "cn=admin,dc=dave3610,dc=com"
```

- Create the following **organizationalUnit** in **OpenLDAP**:
    - **People**
    - **Groups**

```
# Populate db with Organizational Units (OU). This is to categorize ldap
elements to give them Distinguished Name (DN). For managing user accounts we
need a People/Users and Group OU

# Create LDIF file that contains basic Ous
ubuntu@first-instance:~$ nvim base.ldif

dn: ou=People,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: People

dn: ou=Groups,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: Groups

# Add entries of the LDIF file to the LDAP database
ubuntu@first-instance:~$ ldapadd -x -D cn=admin,dc=dave3610,dc=com -W -f
base.ldif
Enter LDAP Password:
adding new entry "ou=People,dc=dave3610,dc=com"

adding new entry "ou=Groups,dc=dave3610,dc=com"

# Query the database again to see if our OU`s are added
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:/// -b dc=dave3610,dc=com
dn: dc=dave3610,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dave3610
dc: dave3610

dn: ou=People,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: People

dn: ou=Groups,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: Groups
```

- Create the following groups in the **organizationalUnit** Groups:
    - **engineering** with GID **4000**
    - **sysadmins** with GID **5000**
    - **marketing** with GID **6000**

```
# Install and use ldapscripts to create users and groups
ubuntu@first-instance:~$ sudo apt install ldapscripts

# Check the config file at /etc/ldapscripts/ldapscripts.conf to see if it
contains correct parameters
ubuntu@first-instance:~$ cat /etc/ldapscripts/ldapscripts.conf


SERVER="localhost"
SASLAUTH=""
SUFFIX="dc=dave3610,dc=com" # Global suffix
GSUFFIX="ou=Groups"         # Groups ou (just under $SUFFIX)
USUFFIX="ou=People"         # Users ou (just under $SUFFIX)
MSUFFIX="ou=Machines"       # Machines ou (just under $SUFFIX)
GIDSTART=10000
UIDSTART=10000
MIDSTART=10000
BINDDN="cn=admin,dc=dave3610,dc=com"
BINDPWDFILE="/etc/ldapscripts/ldapscripts.passwd"
PASSWORDGEN="pwgen"
RECORDPASSWORDS="yes"
PASSWORDFILE="/var/log/ldapscripts_passwd.log"
LOGTOFILE="yes"
LOGFILE="/var/log/ldapscripts.log"
LOGTOSYSLOG="no"
SYSLOGFACILITY="local4"
SYSLOGLEVEL="info"
LDAPSEARCHBIN="/usr/bin/ldapsearch"
LDAPADDBIN="/usr/bin/ldapadd"
LDAPDELETEBIN="/usr/bin/ldapdelete"
LDAPMODIFYBIN="/usr/bin/ldapmodify"
LDAPMODRDNBIN="/usr/bin/ldapmodrdn"
LDAPPASSWDBIN="/usr/bin/ldappasswd"
LDAPSEARCHOPTS="-o ldif-wrap=no"
GETENTPWCMD=""
GETENTGRCMD=""
GTEMPLATE=""
UTEMPLATE="/etc/ldapscripts/ldapadduser.template"
MTEMPLATE=""

ubuntu@first-instance:~$ sudo cp /etc/ldapscripts/ldapscripts.conf
/etc/ldapscripts/ldapscripts.conf.back
ubuntu@first-instance:~$ sudo sh -c 'cat ldapscripts.example >
/etc/ldapscripts/ldapscripts.conf'

# Populate /etc/ldapscripts/ldapscripts.passwd with admin password `pagh3377`
ubuntu@first-instance:~$ sudo sh -c 'echo -n 'pagh3377' >
/etc/ldapscripts/ldapscripts.passwd'

# Change rights so only root can read the passwd file
ubuntu@first-instance:~$ sudo chmod 400 /etc/ldapscripts/ldapscripts.passwd

# Create group sysadmins with ldapscripts to the ldap database to check info
ubuntu@first-instance:~$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b
cn=config dn
```

```
dn: cn=config
dn: cn=module{0},cn=config
dn: cn=schema,cn=config
dn: cn={0}core,cn=schema,cn=config
dn: cn={1}cosine,cn=schema,cn=config
dn: cn={2}nis,cn=schema,cn=config
dn: cn={3}inetorgperson,cn=schema,cn=config
dn: olcDatabase={-1}frontend,cn=config
dn: olcDatabase={0}config,cn=config
dn: olcDatabase={1}mdb,cn=config

~~~

# Because of the password fuckup, ldapaddgroup won't run because of invalid
credentials. So I added the group manually
ubuntu@first-instance:~$ nvim addgroup.ldif
dn: cn=sysadmins,ou=Groups,dc=dave3610,dc=com
objectClass: top
objectClass: posixGroup
cn: sysadmins
gidNumber: 6000

ubuntu@first-instance:~$ ldapadd -x -D "cn=admin,dc=dave3610,dc=com" -W -f
addgroup.ldif

# Query db to see if group is added now
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:/// -b dc=dave3610,dc=com
dn: dc=dave3610,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dave3610
dc: dave3610

dn: ou=People,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: People

dn: ou=Groups,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: Groups

dn: cn=sysadmins,ou=Groups,dc=dave3610,dc=com
objectClass: top
objectClass: posixGroup
cn: sysadmins
gidNumber: 6000

~~~

# Fixed the authentication problem after this, so added the actual groups with
ldapaddgroup

# Removed the sysadmins group
ubuntu@first-instance:~$ ldapdelete -x -D "cn=admin,dc=dave3610,dc=com" -W
"cn=sysadmins,ou=Groups,dc=da>

# Add engineering with GID 4000, sysadmins 5000 and marketing 6000
ubuntu@first-instance:~$ sudo ldapaddgroup engineriing 4000
Successfully added group engineriing to LDAP
ubuntu@first-instance:~$ sudo ldapaddgroup sysadmins 5000
```

```
Successfully added group sysadmins to LDAP
ubuntu@first-instance:~$ sudo ldapaddgroup marketing 6000
Successfully added group marketing to LDAP

# Verify that groups are added
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:/// -b dc=dave3610,dc=com
dn: dc=dave3610,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dave3610
dc: dave3610

dn: cn=admin,dc=dave3610,dc=com
objectClass: simpleSecurityObject
objectClass: organizationalRole
cn: admin
description: LDAP administrator

dn: ou=Groups,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: Groups

dn: ou=People,dc=dave3610,dc=com
objectClass: organizationalUnit
ou: People

dn: cn=marketing,ou=Groups,dc=dave3610,dc=com
objectClass: posixGroup
cn: marketing
gidNumber: 6000
description: Group account

dn: cn=sysadmins,ou=Groups,dc=dave3610,dc=com
objectClass: posixGroup
cn: sysadmins
gidNumber: 5000
description: Group account

dn: cn=engineriing,ou=Groups,dc=dave3610,dc=com
objectClass: posixGroup
cn: engineriing
gidNumber: 4000
description: Group account
```

- Create the following users in the **organizationalUnit** People
    - **ismail** with UID **4000**, GID **4000**, homeDir **/home/ismail**
    - **amir** with UID **5000**, GID **5000**, homeDir **/home/amir**
    - **geir** with UID **6000**, GID **6000**, homeDir **/home/geir**
    - Give each user a random password
        - **NB!** Password must not be hardcoded but generated.

```
# Create a template file for adding user
ubuntu@first-instance:~$ nvim ldapadduser.template

dn: uid=<user>,<usuffix>,<suffix>
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: <user>
sn: <user>
givenName: <user>
displayName: <user>
uid: <user>
uidNumber: <uid>
gidNumber: <gid>
homeDirectory: <home>
loginShell: <shell>
gecos: <user>
mail: <user>@dave3610.com

# Move our template to /etc/ldapscripts/
ubuntu@first-instance:~$ sudo cp ldapadduser.template
/etc/ldapscripts/ldapadduser.template

# Ideally we would use `sudo ldapadduser <name> <group>. However, the
assignment specifies a UID, so then we need to create a LDIF file per user

# I decided to create a script to generate a user LDIF file
ubuntu@first-instance:~$ nvim create_userldif.sh
```
#!/bin/bash

Script to create a user LDIF file and add the user

# Variables
username="$1"
uid="$2"
gid="$3"
home="/home/$username"

# Generate a random password
password=$(pwgen 12 1)

Create user LDIF file
cat > ${username}.ldif <<EOF
dn: uid=$username,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: $username
sn: $username
givenName: $username
displayName: $username
```
```

```
uid: $username
uidNumber: $uid
gidNumber: $gid
homeDirectory: $home
loginShell: /bin/bash
gecos: $username
userPassword: $(slappasswd -h {md5} -s $password)
mail: $username@dave3610.com
EOF

# Add user to LDAP
ldapadd -x -D "cn=admin,dc=dave3610,dc=com" -W -f ${username}.ubuntu@first-
instance:~$ nvim create_userldif.shldif
```

```
ubuntu@first-instance:~$ sudo chmod u+x create_userldif.sh
ubuntu@first-instance:~$ ./create_userldif.sh ismail 4000 4000
Enter LDAP Password:
adding new entry "uid=ismail,ou=People,dc=dave3610,dc=com"

ubuntu@first-instance:~$ ./create_userldif.sh amir 5000 5000
Enter LDAP Password:
adding new entry "uid=amir,ou=People,dc=dave3610,dc=com"

ubuntu@first-instance:~$ ./create_userldif.sh geir 6000 6000
Enter LDAP Password:
adding new entry "uid=geir,ou=People,dc=dave3610,dc=com"

# Verify that users are added
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:/// -b dc=dave3610,dc=com

dn: uid=amir,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: amir
sn: amir
givenName: amir
displayName: amir
uid: amir
uidNumber: 5000
gidNumber: 5000
homeDirectory: /home/amir
loginShell: /bin/bash
gecos: amir
mail: amir@dave3610.com

dn: uid=geir,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: geir
sn: geir
givenName: geir
displayName: geir
uid: geir
uidNumber: 6000
gidNumber: 6000
homeDirectory: /home/geir
loginShell: /bin/bash
gecos: geir
```

```
mail: geir@dave3610.com

dn: uid=ismail,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: ismail
sn: ismail
givenName: ismail
displayName: ismail
uid: ismail
uidNumber: 4000
gidNumber: 4000
homeDirectory: /home/ismail
loginShell: /bin/bash
gecos: ismail
mail: ismail@dave3610.com

# Found out that the password is not logged because of my script. So a new
approach is to delete the users, and use ldapadduser as usual, and then
ldapmodifyuser to change uid after they are created
ubuntu@first-instance:~$ sudo ldapdeleteuser ismail
Successfully deleted user uid=ismail,ou=People,dc=dave3610,dc=com from LDAP
ubuntu@first-instance:~$ sudo ldapdeleteuser geir
Successfully deleted user uid=geir,ou=People,dc=dave3610,dc=com from LDAP
ubuntu@first-instance:~$ sudo ldapdeleteuser amir
Successfully deleted user uid=amir,ou=People,dc=dave3610,dc=com from LDAP

ubuntu@first-instance:~$ sudo ldapadduser ismail 4000
Successfully added user ismail to LDAP
Successfully set password for user ism
ubuntu@first-instance:~$ sudo ldapadduser amir 5000
Successfully added user amir to LDAP
Successfully set password for user amir
ubuntu@first-instance:~$ sudo ldapadduser geir 6000
Successfully added user geir to LDAP
Successfully set password for user geir

ubuntu@first-instance:~$ sudo ldapmodifyuser ismail
# About to modify the following entry :
# Enter your modifications here, end with CTRL-D.
dn: uid=ismail,ou=People,dc=dave3610,dc=com
replace: uidNumber
uidNumber: 4000
Successfully modified user entry uid=ismail,ou=People,dc=dave3610,dc=com in
LDAP

ubuntu@first-instance:~$ sudo ldapmodifyuser amir
# About to modify the following entry :
# Enter your modifications here, end with CTRL-D.
dn: uid=amir,ou=People,dc=dave3610,dc=com
replace: uidNumber
uidNumber: 5000
Successfully modified user entry uid=amir,ou=People,dc=dave3610,dc=com in LDAP

ubuntu@first-instance:~$ sudo ldapmodifyuser geir
# About to modify the following entry :
# Enter your modifications here, end with CTRL-D.
dn: uid=geir,ou=People,dc=dave3610,dc=com
replace: uidNumber
uidNumber: 6000
```

```
Successfully modified user entry uid=geir,ou=People,dc=dave3610,dc=com in LDAP

# Find password
ubuntu@first-instance:~$ cat /var/log/ldapscripts_passwd.log
uid=ismail,ou=People,dc=dave3610,dc=com : TeiKe8wo
uid=amir,ou=People,dc=dave3610,dc=com : aeph1Ohg
uid=geir,ou=People,dc=dave3610,dc=com : Ka0eebae
```

- Setup an **LDAP** client and make sure that it uses:
  - suffix: dc=dave3610,dc=com
  - rootdn: cn=admin,dc=dave3610,dc=com
  - rootpw: pagh3377 kryptert

```
# I have setup a new server instance for the LDAP client
# First check passwd to see if we have any LDAP users
ubuntu@ldap-client:~$ getent passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System
(admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network
Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time
Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:113::/run/uuidd:/usr/sbin/nologin
tcpdump:x:108:114::/nonexistent:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:110:1::/var/cache/pollinate:/bin/false
landscape:x:111:116::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:117:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false

# Check where databases live on the system
ubuntu@ldap-client:~$ cat /etc/nsswitch.conf
# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch functionality.
```

```
# If you have the `glibc-doc-reference' and `info' packages installed, try:
# `info libc "Name Service Switch"' for information about this file.

passwd:         files systemd
group:          files systemd
shadow:         files
gshadow:        files

hosts:          files dns
networks:       files

protocols:      db files
services:       db files
ethers:         db files
rpc:            db files

netgroup:       nis

# Install libnss-ldap
ubuntu@ldap-client:~$ sudo apt install -y libnss-ldap

# Add ip for LDAP server
10.196.36.97

# Add suffix
dc=dave3610,dc=com

# Select LDAP version
3

# Make local root Database admin
yes

# Require login
No

# LDAP account for root
cn=admin,dc=dave3610.dc=com

# Root password
pagh3377

# Print the config for ldap client (using grep to only print uncommented lines)
ubuntu@ldap-client:~$ cat /etc/ldap.conf |grep -v '#' |grep -v -e '^$'
base dc=dave3610,dc=com
uri ldapi:///10.196.36.97
ldap_version 3
rootbinddn cn=admin,dc=dave3610,dc=net
pam_password md5

# Configure LDAP profile for nsswitch. Can be done manually by editing
/etc/nsswitch.conf, but we are using a tool for doing it automatically
ubuntu@ldap-client:~$ sudo auth-client-config -t nss -p lac_ldap
sudo: auth-client-config: command not found

# Check the content of nsswitch.conf if it is edited
ubuntu@ldap-client:~$ cat /etc/nsswitch.conf
# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch functionality.
# If you have the `glibc-doc-reference' and `info' packages installed, try:
```

```
# `info libc "Name Service Switch"' for information about this file.

passwd:         files systemd
group:          files systemd
shadow:         files
gshadow:        files

hosts:          files dns
networks:       files

protocols:      db files
services:       db files
ethers:         db files
rpc:            db files

netgroup:       nis
```

 # It has not, so I'll have to do a manual edit

```
ubuntu@ldap-client:~$ sudo nvim /etc/nsswitch.conf

# /etc/nsswitch.conf
#
# Example configuration of GNU Name Service Switch functionality.
# If you have the `glibc-doc-reference' and `info' packages installed, try:
# `info libc "Name Service Switch"' for information about this file.

passwd:         files ldap
group:          files ldap
shadow:         files ldap
gshadow:        files

hosts:          files dns
networks:       files

protocols:      db files
services:       db files
ethers:         db files
rpc:            db files

netgroup:       nis
```

 # Configure PAM to use ldap. PAM are usually found at /etc/pam.d/

```
ubuntu@ldap-client:~$ cat /etc/pam.d/common-* |grep -v '#'

account     [success=2 new_authtok_reqd=done default=ignore]     pam_unix.so
account     [success=1 default=ignore]    pam_ldap.so
account     requisite               pam_deny.so
account     required                pam_permit.so

auth  [success=2 default=ignore]    pam_unix.so nullok
auth  [success=1 default=ignore]    pam_ldap.so use_first_pass
auth  requisite               pam_deny.so
auth  required                pam_permit.so
auth  optional                pam_cap.so



password    [success=2 default=ignore]    pam_unix.so obscure yescrypt
password    [success=1 user_unknown=ignore default=die]     pam_ldap.so
use_authtok try_first_pass
```

```
password    requisite                    pam_deny.so
password    required                     pam_permit.so

session     [default=1]                  pam_permit.so
session     requisite                    pam_deny.so
session     required                     pam_permit.so
session optional                  pam_umask.so
session     required    pam_unix.so
session     optional                     pam_ldap.so
session     optional    pam_systemd.so

session     [default=1]                  pam_permit.so
session     requisite                    pam_deny.so
session     required                     pam_permit.so
session optional                  pam_umask.so
session     required    pam_unix.so
session     optional                     pam_ldap.so
```

 # PAM uses the pam_unix.so module. We need to configure it to use LDAP for authentication. We can manually edit each files under /etc/pam.d/ directory or use the tool pam-auth-update

```
ubuntu@ldap-client:~$ sudo pam-auth-update
```

 # PAM profiles to enable

```
[*] Unix authentication
[*] LDAP Authentication
[*] Register user sessions in the systemd control group hierarchy
[*] Create home directory on login
[*] Inheritable Capabilities Management
```

 # Check the PAM files again

```
ubuntu@ldap-client:~$ cat /etc/pam.d/common-* |grep -v '#'

account     [success=2 new_authtok_reqd=done default=ignore]     pam_unix.so
account     [success=1 default=ignore]    pam_ldap.so
account     requisite                    pam_deny.so
account     required                     pam_permit.so

auth  [success=2 default=ignore]    pam_unix.so nullok
auth  [success=1 default=ignore]    pam_ldap.so use_first_pass
auth  requisite                   pam_deny.so
auth  required                    pam_permit.so
auth  optional                    pam_cap.so



password    [success=2 default=ignore]    pam_unix.so obscure yescrypt
password    [success=1 user_unknown=ignore default=die]    pam_ldap.so
use_authtok try_first_pass
password    requisite                    pam_deny.so
password    required                     pam_permit.so

session     [default=1]                  pam_permit.so
session     requisite                    pam_deny.so
session     required                     pam_permit.so
session optional                  pam_umask.so
session     required    pam_unix.so
session     optional                     pam_ldap.so
session     optional    pam_systemd.so
session     optional                     pam_mkhomedir.so
```

```
session     [default=1]             pam_permit.so
session     requisite               pam_deny.so
session     required                pam_permit.so
session optional              pam_umask.so
session     required    pam_unix.so
session     optional                pam_ldap.so

#  PAM now uses pam_unix.so and pam_ldap.so modules

# Check passwd to see if LDAP users are there
ubuntu@ldap-client:~$ getent passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System
(admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network
Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time
Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:113::/run/uuidd:/usr/sbin/nologin
tcpdump:x:108:114::/nonexistent:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:110:1::/var/cache/pollinate:/bin/false
landscape:x:111:116::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:117:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false

# No users. Install ldap-utils and test ldapsearch for the server
ubuntu@ldap-client:~$ ldapsearch -x -LLL -H ldap://10.196.36.97 -b
"ou=People,dc=dave3610,dc=com" "(objectClass=inetOrgPerson)"

dn: uid=ismail,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: ismail
sn: ismail
[…]
```

```
# Tried to install nslcd to help with configuration, but it dit not work. Saw
the steps in the note and tried following them on both server and client
ubuntu@ldap-client:~$ cp /etc/ssh/sshd_config /etc/ssh/sshd_config.backup
ubuntu@ldap-client:~$ sed -i 's/PasswordAuthentication
no/PasswordAuthentication yes/g' /etc/ssh/sshd_config
ubuntu@ldap-client:~$ sudo systemctl restart sshd

# Try getent passwd again
ubuntu@ldap-client:~$ getent passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System
(admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network
Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time
Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:113::/run/uuidd:/usr/sbin/nologin
tcpdump:x:108:114::/nonexistent:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:110:1::/var/cache/pollinate:/bin/false
landscape:x:111:116::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:117:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false

# Still nothing. I give up.
```

Note that

OpenStack instances are provisioned with PasswordAuthentication no in /etc/ssh/sshd_config, causing we cannot ssh with LDAP user credentials to the LDAP client instance. We must set PasswordAuthentication to yes and then restart the sshd service to enable tunnelled clear text passwords.

- cp /etc/ssh/sshd_config /etc/ssh/sshd_config.backup
- sed -i 's/PasswordAuthentication no/PasswordAuthentication yes/g' /etc/ssh/sshd_config
- sudo systemctl restart sshd

Assignment-7: Managing permission.

Objectives: The goal of this week's assignment is to:

- learn to manage **permissions** in Linux
- learn to manage **Quotas**
- know how to delegate root privileges to users with **sudo**

Hi junior Sysadmin,

It is great that in your last task/assignment, you managed to create, disable and delete users with the built-in Linux utilities. In this week, we would like you to explore and get accustomed to Linux permission, disk resource management with Quotas and delegate root privileges.

Some useful resources to start with are:

- **Chapter 3** and **Chapter 7.4** of Linux System Administration I by Tuxcademy
    - https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
- **Chapter 32, 33** and **34** of Linux System Administration by Paul Cobbaut
    - http://linux-training.be/funhtml/pt09.html

Best regards

Senior Sysadmin!.

Step 1: Managing Permissions:

In **Assignment-5**, you created several users and groups. You will use those users and groups in the following tasks.

The home directories for all the users, created in the previous assignment, must only be accessible to their **owner** and their **primary group** only. For example, **/home/luke** should only be accessible to the owner **luke** and his primary group **rebels.** **NB!**. This will require that you remove the user **kirk** from his secondary group **rebels**

Your task as a sysadmin is to apply the necessary permissions according to the requirements stated above!

```
# Check which groups kirk is a part of
ubuntu@first-instance:~$ id kirk
uid=1001(kirk) gid=1001(captains) groups=1001(captains),1003(rebels)

# Remove kirk from group rebels
ubuntu@first-instance:~$ sudo gpasswd --delete kirk rebels

# Check if kirk is removed from rebels
ubuntu@first-instance:~$ id kirk
uid=1001(kirk) gid=1001(captains) groups=1001(captains)

# Create a script for setting chmod permissions
ubuntu@first-instance:~$ nvim set_home.sh
```
```
#! /bin/bash

# Basic script for setting full rights for user and primary group on their home
directory

# Loop through every home directory
for home in /home/*
do
      # Check if the home is a directory that exists
      if [ -d "$home" ]
      then
            echo "Setting permission for $home"
            chmod 770 "$home"
      fi
done
```

```
# Set exectute permission on the script
ubuntu@first-instance:~$ chmod u+x set_home.sh

# Run script
ubuntu@first-instance:~$ sudo ./set_home.sh
Setting permission for /home/bart
Setting permission for /home/kirk
Setting permission for /home/luke
Setting permission for /home/ubuntu
Setting permission for /home/user1
Setting permission for /home/user2
Setting permission for /home/user3
Setting permission for /home/user4
Setting permission for /home/user5
```

```
Setting permission for /home/user6
Setting permission for /home/user7
```

The users **user2** and **user4** want to work on a project called **project-1** together so that both of them have access to the files (**read**, **write** and **execute** permissions), but do not want other users or anyone else other than root to have access to those files.

1. Neither **user2** and **user4** should have access to each other files other than the files they are working on that belongs to the project.
2. Create a directory named **projects** under the root / directory. Then create a directory named **project-1** under **/projects** so that your final directory path is **/projects/project-1**
3. Your task as a sysadmin is to apply the necessary permissions on the newly created directory according to the requirements stated above!

```
# I am assuming that user2 and user4 should be able to access all files in
project-1, but not their personal files in home. I assume they are supposed to
have full collaborative access to all contents.

# Create a project group and add members
ubuntu@first-instance:~$ sudo groupadd project1-group
ubuntu@first-instance:~$ sudo usermod -aG project1-group user2
ubuntu@first-instance:~$ sudo usermod -aG project1-group user4

# Create projects directories
ubuntu@first-instance:~$ sudo mkdir -p /projects/project-1
ubuntu@first-instance:~$ ls /projects
project-1

# Change ownership on the project-1 directory to project1-group
# I also give root access to this directory
ubuntu@first-instance:~$ sudo chown root:project1-group /projects/project-1/

# Set permissions
ubuntu@first-instance:~$ sudo chmod 2770 /projects/project-1
# setgid 2 so all files and directories inherit the group id and 0 to ensure no
one else have permissions

# Check permissions
ubuntu@first-instance:~$ ls -ld /projects/project-1/
drwxrws--- 2 root project1-group 4096 Apr 30 09:23 /projects/project-1/
```

Now, **user4**, **user5** and **user6** want to work on a project called **project-2** together but would like to have some fine-grained permissions as follows:

NB!
This section will require you to setup and use Linux Access Control List (ACL).

1. **user4** must have **read**, **write** and **execute** access
2. **user5** must have **write** and **execute** access
   A. This will give **user5** the possibility to create files and directories, read files, but the **user5** will not be able to list the content of the **project-2** directory.
3. **user6** must have only **read** and **execute** access
4. Everyone else **must not** have any access
5. In addition, giving **read and write access** to the group **rebels**
6. Create a directory named **project-2** under **/projects** so that your final directory path is **/projects/project-2**
7. Your task as a sysadmin is to apply the necessary permissions to the newly created directory according to the requirements stated above!

```
# Install ACL to work with permissions
ubuntu@first-instance:~$ sudo apt install acl

# Create project-2 directory
ubuntu@first-instance:~$ sudo mkdir /projects/project-2

# Set restrictive ownership and permissions to prevent default access
ubuntu@first-instance:~$ sudo chown root /projects/project-2
ubuntu@first-instance:~$ sudo chmod 0700 /projects/project-2

# Using the -d flag to set permission so all files in the directory inherit the
rights

# Set rwx access for user4
ubuntu@first-instance:~$ sudo setfacl -d -m u:user4:rwx /projects/project-2

# Set wx access for user5
ubuntu@first-instance:~$ sudo setfacl -d -m u:user5:wx /projects/project-2

# Set rx access for user6
ubuntu@first-instance:~$ sudo setfacl -d -m u:user6:rx /projects/project-2

# Set rw access to group rebels
ubuntu@first-instance:~$ sudo setfacl -d -m g:rebels:rw /projects/project-2

# Set no access for others
ubuntu@first-instance:~$ sudo setfacl -d -m o::--- /projects/project-2

# Check permissions for the directory to verify
ubuntu@first-instance:~$ getfacl /projects/project-2
getfacl: Removing leading '/' from absolute path names
# file: projects/project-2
# owner: root
# group: root
```

```
user::rwx
user:user4:rwx
user:user5:-wx
user:user6:r-x
group::---
group:rebels:rw-
mask::rwx
other::---
default:user::rwx
default:user:user4:rwx
default:user:user5:-wx
default:user:user6:r-x
default:group::---
default:group:rebels:rw-
default:mask::rwx
default:other::---
```

Explore further Linux permissions by practising the exercises on chapter **32**, **33** and **34** of Linux System Administration by Paul Cobbaut

http://linux-training.be/funhtml/ch32.html#idp55982752

Chapter 32

1. As normal user, create a directory ~/permissions. Create a file owned by yourself in there.

```
ubuntu@first-instance:~$ mkdir ~/permissions
ubuntu@first-instance:~$ touch ~/permissions/myfile
ubuntu@first-instance:~$ ls -l ~/permissions/
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 May  1 08:39 myfile
```

2. Copy a file owned by root from /etc/ to your permissions dir, who owns this file now ?

```
ubuntu@first-instance:~$ ls -alt /etc
ubuntu@first-instance:~$ cp /etc/group ~/permissions/
ubuntu@first-instance:~$ ls -alt ~/permissions/
total 12
drwxrwxr-x 2 ubuntu ubuntu 4096 May  1 08:42 .
-rw-r--r-- 1 ubuntu ubuntu 1050 May  1 08:42 group
-rw-rw-r-- 1 ubuntu ubuntu    0 May  1 08:39 myfile
```

3. As root, create a file in the users ~/permissions directory.

```
ubuntu@first-instance:~$ sudo touch ~/permissions/rootfile
```

4. As normal user, look at who owns this file created by root.

```
ubuntu@first-instance:~$ ls -alt ~/permissions/
total 12
drwxrwxr-x 2 ubuntu ubuntu 4096 May  1 08:44 .
-rw-r--r-- 1 root   root      0 May  1 08:44 rootfile
```

5. Change the ownership of all files in ~/permissions to yourself.

```
ubuntu@first-instance:~$ sudo chown ubuntu ~/permissions/*
ubuntu@first-instance:~$ ls -alt ~/permissions/
total 12
drwxrwxr-x 2 ubuntu ubuntu 4096 May  1 08:44 .
-rw-r--r-- 1 ubuntu root      0 May  1 08:44 rootfile
-rw-r--r-- 1 ubuntu ubuntu 1050 May  1 08:42 group
-rw-rw-r-- 1 ubuntu ubuntu    0 May  1 08:39 myfile
```

6. Make sure you have all rights to these files, and others can only read.

```
ubuntu@first-instance:~$ chmod u+rwx,g=---,o=--- ~/permissions/*
ubuntu@first-instance:~$ ls -alt ~/permissions/
-rwx------ 1 ubuntu root      0 May  1 08:44 rootfile
-rwx------ 1 ubuntu ubuntu 1050 May  1 08:42 group
```

```
-rwx------ 1 ubuntu ubuntu    0 May  1 08:39 myfile
```

7. With chmod, is 770 the same as rwxrwx--- ?

```
ubuntu@first-instance:~$ chmod 770 ~/permissions/rootfile
ubuntu@first-instance:~$ chmod u=rwx,g=rwx,o=--- ~/permissions/group
ubuntu@first-instance:~$ ls -alt ~/permissions/
-rwxrwx--- 1 ubuntu root      0 May  1 08:44 rootfile
-rwxrwx--- 1 ubuntu ubuntu 1050 May  1 08:42 group

# Yes
```

8. With chmod, is 664 the same as r-xr-xr-- ?

```
ubuntu@first-instance:~$ chmod 644 ~/permissions/rootfile
ubuntu@first-instance:~$ chmod u=r-x,g=r-x,o=r-- ~/permissions/group
ubuntu@first-instance:~$ ls -alt ~/permissions/
-rw-r--r-- 1 ubuntu root      0 May  1 08:44 rootfile
-r--r--r-- 1 ubuntu ubuntu 1050 May  1 08:42 group

# No
```

9. With chmod, is 400 the same as r-------- ?

```
ubuntu@first-instance:~$ chmod 400 ~/permissions/rootfile
ubuntu@first-instance:~$ chmod u=r--,g=---,o=--- ~/permissions/group
ubuntu@first-instance:~$ ls -alt ~/permissions/
-r-------- 1 ubuntu root      0 May  1 08:44 rootfile
-r-------- 1 ubuntu ubuntu 1050 May  1 08:42 group

# Yes
```

10. With chmod, is 734 the same as rwxr-xr-- ?

```
ubuntu@first-instance:~$ chmod 734 ~/permissions/rootfile
ubuntu@first-instance:~$ chmod u=rwx,g=r-x,o=r-- ~/permissions/group
ubuntu@first-instance:~$ ls -alt ~/permissions/
-rwx-wxr-- 1 ubuntu root      0 May  1 08:44 rootfile
-rwxr--r-- 1 ubuntu ubuntu 1050 May  1 08:42 group

# No
```

11a. Display the umask in octal and in symbolic form.

```
ubuntu@first-instance:~$ umask
0002
ubuntu@first-instance:~$ umask -S
u=rwx,g=rwx,o=rx
```

11b. Set the umask to 077, but use the symbolic format to set it. Verify that this works.

```
ubuntu@first-instance:~$ umask u=rwx,g=,o=
ubuntu@first-instance:~$ umask -S
```

```
u=rwx,g=,o=
ubuntu@first-instance:~$ umask
0077
```

12. Create a file as root, give only read to others. Can a normal user read this file ? Test writing to this file with vi.

```
ubuntu@first-instance:~$ sudo touch ~/permissions/root2
ubuntu@first-instance:~$ ls -alt ~/permissions/
-rw-r--r-- 1 root   root      0 May  1 09:02 root2
ubuntu@first-instance:~$ vi ~/permissions/root2

# It opens, but I can't write any changes
```

13a. Create a file as normal user, give only read to others. Can another normal user read this file ? Test writing to this file with vi.

```
ubuntu@first-instance:~$ sudo -u kirk touch /home/kirk/kirkfile
ubuntu@first-instance:~$ sudo -u kirk ls -alt /home/kirk
-rw-r--r--  1 kirk captains    0 May  1 09:18 kirkfile
ubuntu@first-instance:~$ sudo -u bart vi /home/kirk/kirkfile

# Can open and read the file, but no write permissions
```

13b. Can root read this file ? Can root write to this file with vi ?

```
ubuntu@first-instance:~$ sudo vi /home/kirk/kirkfile
ubuntu@first-instance:~$ sudo cat /home/kirk/kirkfile
poop

# Yes
```

14. Create a directory that belongs to a group, where every member of that group can read and write to files, and create files. Make sure that people can only delete their own files.

```
# Create a project group and add members
ubuntu@first-instance:~$ sudo groupadd group32
ubuntu@first-instance:~$ sudo usermod -aG group32 kirk
ubuntu@first-instance:~$ sudo usermod -aG group32 bart
ubuntu@first-instance:~$ sudo usermod -aG group32 user2

# Create projects directories
ubuntu@first-instance:~$ sudo mkdir /projects/project32

# Change ownership on the project32 directory to group32
# I also give root access to this directory
ubuntu@first-instance:~$ sudo chown root:group32 /projects/project32/

# Set permissions. Set 3 to get setgid (recursive) and sticky bit
ubuntu@first-instance:~$ sudo chmod 3770 /projects/project32/

# Check permissions
ubuntu@first-instance:~$ ls -ld /projects/project32
drwxrws--T 2 root group32 4096 May  1 09:34 /projects/project32
```

Chapter 33

1a. Set up a directory, owned by the group sports.

```
ubuntu@first-instance:~$ sudo groupadd sports
ubuntu@first-instance:~$ mkdir ~/permissions/sports
ubuntu@first-instance:~$ sudo chown root:sports ~/permissions/sports
ubuntu@first-instance:~$ ls -alt ~/permissions/
drwxrwxr-x 2 root   sports 4096 May  1 10:59 sports
```

1b. Members of the sports group should be able to create files in this directory.

```
# They already have write permission, but just to make sure:
ubuntu@first-instance:~$ sudo chmod 770 ~/permissions/sports/
ubuntu@first-instance:~$ ls -alt ~/permissions/
drwxrwx--- 2 root   sports 4096 May  1 10:59 sports
```

1c. All files created in this directory should be group-owned by the sports group.

```
# setgid bit (2)
ubuntu@first-instance:~$ sudo chmod 2770 ~/permissions/sports/
ubuntu@first-instance:~$ ls -alt ~/permissions/
drwxrws--- 2 root   sports 4096 May  1 10:59 sports
```

1d. Users should be able to delete only their own user-owned files.

```
# setgid and sticky bit (3)
ubuntu@first-instance:~$ sudo chmod 3770 ~/permissions/sports/
ubuntu@first-instance:~$ ls -alt ~/permissions/
drwxrws--T 2 root   sports 4096 May  1 10:59 sports
```

1e. Test that this works!

```
# Add users to group
ubuntu@first-instance:~$ sudo usermod -aG sports user1
ubuntu@first-instance:~$ sudo usermod -aG sports user2

# Switch to users and create files in the sports directory
ubuntu@first-instance:~$ sudo -u user1 touch /projects/sports/user1file
ubuntu@first-instance:~$ sudo -u user2 touch /projects/sports/user2file

# Try to delete files as user1
ubuntu@first-instance:~$ sudo -u user1 rm /projects/sports/user1file
ubuntu@first-instance:~$ sudo -u user1 rm /projects/sports/user2file
rm: remove write-protected regular empty file '/projects/sports/user2file'? y
rm: cannot remove '/projects/sports/user2file': Operation not permitted
ubuntu@first-instance:~$ sudo -u user1 ls -alt /projects/sports
-rw-r--r-- 1 user2 sports    0 May  1 11:10 user2file

# Create a new file as user2 and check ownership
ubuntu@first-instance:~$ sudo -u user2 touch /projects/sports/newfile
ubuntu@first-instance:~$ sudo -u user2 ls -l /projects/sports/
-rw-r--r-- 1 user2 sports 0 May  1 11:13 newfile
```

2. Verify the permissions on /usr/bin/passwd. Remove the setuid, then try changing your password as a normal user. Reset the permissions back and try again.

```
# Verify permissions
ubuntu@first-instance:~$ ls -l /usr/bin/passwd
-rwsr-xr-x 1 root root 59976 Feb  6 12:54 /usr/bin/passwd # setuid bit set

# Remove setuid bit
ubuntu@first-instance:~$ sudo chmod u-s /usr/bin/passwd

# Verify removal
ubuntu@first-instance:~$ ls -l /usr/bin/passwd
-rwxr-xr-x 1 root root 59976 Feb  6 12:54 /usr/bin/passwd # no s for setuid

# Test changing password as normal user
ubuntu@first-instance:~$ sudo -u user1 passwd
Changing password for user1.
Current password:
passwd: Authentication token manipulation error
passwd: password unchanged

# Revert and check that setuid is back
ubuntu@first-instance:~$ sudo chmod u+s /usr/bin/passwd
ubuntu@first-instance:~$ ls -l /usr/bin/passwd
-rwsr-xr-x 1 root root 59976 Feb  6 12:54 /usr/bin/passwd # s is back
```

3. If time permits (or if you are waiting for other students to finish this practice), read about file attributes in the man page of chattr and lsattr. Try setting the i attribute on a file and test that it works.

```
# Create file and make it immutable
ubuntu@first-instance:~$ touch ~/permissions/ifile
ubuntu@first-instance:~$ sudo chattr +i ~/permissions/ifile
ubuntu@first-instance:~$ lsattr ~/permissions/ifile
----i---------e------- /home/ubuntu/permissions/ifile

# Test immutability
ubuntu@first-instance:~$ echo "poop" > ~/permissions/ifile
-bash: /home/ubuntu/permissions/ifile: Operation not permitted
ubuntu@first-instance:~$ rm ~/permissions/ifile
rm: cannot remove '/home/ubuntu/permissions/ifile': Operation not permitted
ubuntu@first-instance:~$ mv ~/permissions/ifile newfile
mv: cannot move '/home/ubuntu/permissions/ifile' to 'newfile': Operation not
permitted

# Remove immutability and test
ubuntu@first-instance:~$ sudo chattr -i ~/permissions/ifile
ubuntu@first-instance:~$ lsattr ~/permissions/ifile
--------------e------- /home/ubuntu/permissions/ifile
ubuntu@first-instance:~$ echo "poop" > ~/permissions/ifile
ubuntu@first-instance:~$ cat ~/permissions/ifile
poop
```

Step 2 - Managing quotas

In this section we want you to set up disk **quotas** for our users.

1. Disk quotas are set up on a per file system basis. **Quotas** may be set for individual user accounts and/or groups. We are going to set up quotas for user accounts in the file system
2. A useful resource on this section is:
   A. https://www.ibm.com/developerworks/library/l-lpic1-104-4/index.html
3. **NB!** You will need to install the **Quota** package and configure it before it can be set or used on the system
4. **NB!** if "modprobe quota_v1" or "modprobe quota_v2" return some errors. Please install a "linux-modules-extra-$(uname -r)" package too.

```
# Install quota
ubuntu@first-instance:~$ sudo apt install quota quotatool
ubuntu@first-instance:~$ sudo apt install linux-modules-extra-$(uname -r) #
just to be safe

# First enable quota on partition by adding usrquota and grpquota in fstab
ubuntu@first-instance:~$ sudo nvim /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-
ro,usrquota,grpquota     0 1

# Reboot for changes to take effect (and check that server do not crash)
ubuntu@first-instance:~$ sudo reboot

# Initialize quota database and turn on quota
ubuntu@first-instance:~$ sudo quotacheck -ugm /
ubuntu@first-instance:~$ sudo quotacheck -ugm /
ubuntu@first-instance:~$ sudo quotaon -ugv /
quotaon: Your kernel probably supports ext4 quota feature but you are using
external quota files. Please switch your filesystem to use ext4 quota feature
as external quota files on ext4 are deprecated.
/dev/sda1 [/]: group quotas turned on
/dev/sda1 [/]: user quotas turned on

# Check quotas set
ubuntu@first-instance:~$ sudo repquota -s /
*** Report for user quotas on device /dev/sda1
Block grace time: 7days; Inode grace time: 7days
                        Space limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --   3087M      0K      0K           121k     0     0
[…]
ubuntu    --    112M      0K      0K          16695     0     0
kirk      --     20K      0K      0K              5     0     0
bart      --     20K      0K      0K              5     0     0
luke      --     16K      0K      0K              4     0     0
user1     --      4K      0K      0K              1     0     0
user2     --      4K      0K      0K              3     0     0
user3     --      4K      0K      0K              1     0     0
user4     --      4K      0K      0K              1     0     0
user5     --      4K      0K      0K              1     0     0
user6     --      4K      0K      0K              1     0     0
user7     --      4K      0K      0K              1     0     0
[…]
```

5. Setup Quota that limits the following users disk space:
   A. **user2** soft limit to **80 MB** and the hard limit to **100 MB**
   B. **user3** soft limit to **40 MB** and the hard limit to **50 MB**
   C. **user4** soft limit to **10 MB** and the hard limit to **20 MB**
   D. **user5** soft limit to **5 MB** and the hard limit to **10 MB**

```
# Set soft limit 80MB and hard limit 100MB for user2
# First use numfmt to convert MB to KB used by quota
ubuntu@first-instance:~$ numfmt --from=auto 80Mi --to-unit=1000; numfmt --
from=auto 100Mi --to-unit=1000
83887
104858

ubuntu@first-instance:~$ sudo edquota -u user2

Disk quotas for user user3 (uid 6003):
Filesystem          blocks       soft       hard     inodes     soft     hard
/dev/sda1                4          83887     104858      1         0        0


# Set soft limit 40MB and hard limit 50MB for user3
ubuntu@first-instance:~$ numfmt --from=auto 40Mi --to-unit=1000; numfmt --
from=auto 50Mi –to-unit=1000
41944
52429

ubuntu@first-instance:~$ sudo edquota -u user3

Disk quotas for user user3 (uid 6003):
Filesystem           blocks      soft       hard     inodes     soft     hard
/dev/sda1                4      41944      52429          1        0        0

# Set soft limit 10MB and hard limit 20MB for user4
ubuntu@first-instance:~$ numfmt --from=auto 10Mi --to-unit=1000; numfmt --
from=auto 20Mi --to-unit=1000
10486
20972

ubuntu@first-instance:~$ sudo edquota -u user4

Disk quotas for user user4 (uid 6004):
Filesystem           blocks       soft       hard     inodes     soft     hard
/dev/sda1                 4        10486    20972          1        0        0


# Set soft limit 5MB and hard limit 10MB for user5
ubuntu@first-instance:~$ numfmt --from=auto 5Mi --to-unit=1000; numfmt --
from=auto 10Mi --to-unit=1000
5243
10486

ubuntu@first-instance:~$ sudo edquota -u user5

Disk quotas for user user5 (uid 6005):
Filesystem          blocks       soft       hard     inodes     soft     hard
/dev/sda1                4       5243      10486          1        0        0
```

6. **Test that Quota is working by:**
   A. Login as user **user5**
   B. Use the **dd** command to copy data in Mb null bytes to a file.

      dd if=/dev/zero of=/tmp/dd-data-file count=100 bs=1M

   C. If you get the warning "**Disk quota exceeded**", then the quota system is working properly otherwise troubleshoot the problem and make it work

```
ubuntu@first-instance:~$ sudo -u user5 dd if=/dev/zero of=/tmp/dd-data-file
count=100 bs=1M
dd: error writing '/tmp/dd-data-file': Disk quota exceeded
10+0 records in
9+0 records out
10481664 bytes (10 MB, 10 MiB) copied, 0.0253128 s, 414 MB/s
```

Step 3 - Delegating root privileges through sudo

1. Change to the user **user2**. This can be done with the command:
   su user2
2. Try to shutdown the system by executing the following command with **sudo**:
   sudo poweroff
3. You should get the following warning:
   user2 is not in the sudoers file. This incident will be reported.
4. Give **user2** the ability to **shutdown** or **reboot** the system through **sudo**.

   A. **sudo** must not ask **user2** for his or her password when executing the command.

   B. Your task as a sysadmin is to grant this to **user2**

```
# Login to user2, then try shutdown
ubuntu@first-instance:~$ su user2
Password: ****
$ sudo poweroff
[sudo] password for user2: ****
user2 is not in the sudoers file.  This incident will be reported.
$ exit

# Add user2 to the sudoers file with access to poweroff and reboot without
password
ubuntu@first-instance:~$ sudo visudo
user2 ALL=(ALL) NOPASSWD: /sbin/poweroff, /sbin/reboot

# Try to reboot as user2
ubuntu@first-instance:~$ su user2
Password: ****
$ sudo reboot
Connection to 10.196.36.97 closed by remote host.
Connection to 10.196.36.97 closed.
```

Assignment-8: Centralized Home Directories through NFS

The goal in this assignment is to continue where we left off in last weeks assignment and set up an NFS server that will export the home directories of the LDAP users.

Hi Junior Sysadmin,

Great job in setting up the **LDAP** server and creating the accounts for the employees last week. It turns out that the new accounts do not have a centralized **Home Directory**. This will create an unnecessary inconvenience for the users as their files will be spread across many machines and will be cumbersome for them to access.

Set up an **NFS** server that exports the home directories of the LDAP users to any client/machine that is on the Local Area Network (LAN) **192.168.10.0/24**. Take the following requirements on the next page into consideration when setting up the NFS

Note that: since we are on the **OpenStack Cloud**, it has its IP range. You have to adapt it to your configuration.

Server: 10.196.36.97

Client: 10.196.36.113

Best regards
Senior sysadmin!

1. The **NFS** server must be on the same machine as the **LDAP** server

```
ubuntu@first-instance:~$ sudo apt install -y nfs-kernel-server

# Install nfs on client
ubuntu@ldap-client:~$ sudo apt install -y nfs-common
```

2. Create the directory **/NFS/home** on the NFS server

```
ubuntu@first-instance:~$ sudo mkdir -p /NFS/home
```

3. The home directories for any new LDAP users must be created on **/NFS/home** directory

```
ubuntu@first-instance:~$ sudo mkdir -p /NFS/home/ismail
ubuntu@first-instance:~$ sudo mkdir -p /NFS/home/geir
ubuntu@first-instance:~$ sudo mkdir -p /NFS/home/amir
ubuntu@first-instance:~$ ls /NFS/home/
amir  geir  ismail
```

4. The **NFS** server must export the **/NFS/home** directory to all clients/machines on the LAN **192.168.10.0/24** with the **options (rw,sync,no_subtree_check)**

```
# Edit exports to configure NFS server to export /NFS/home directory to
localhost
# I use * as ip, since it is fuckery with OpenStack
ubuntu@first-instance:~$ sudo nvim /etc/exports

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
/NFS/home *(rw,sync,no_subtree_check)

# Make NFS server aware of the changes and restart server
ubuntu@first-instance:~$ sudo exportfs -a
ubuntu@first-instance:~$ sudo service nfs-kernel-server restart

# Use nfsstat to get info about NFS
ubuntu@first-instance:~$ nfsstat -m
# Returns nothing because nothing is mounted
```

5. Any client/machine that will mount the **NFS** share/folder from the **NFS** server must not mount the share/folder permanently (Statically). The client must utilize a mechanism such as **autofs** to mount the share/folder upon users **login** and eventually unmount the share after a certain period of time configured by the sysadmin.

```
# Check if autofs is installed on ldap client, and if not install and start it
ubuntu@ldap-client:~$ dpkg -l | grep autofs
ubuntu@ldap-client:~$ sudo apt install autofs
ubuntu@ldap-client:~$ sudo systemctl start autofs
ubuntu@ldap-client:~$ sudo systemctl enable autofs

# Add config to the auto.master file
ubuntu@ldap-client:~$ sudo nvim /etc/auto.master

# Specify automount with timeout and adds ghost directories for when it is not
mounted
/home   /etc/auto.home --timeout=600 –-ghost

# Create auto.home for config for the home directory. Added timeo and retrans
```

```
so it won't try to connect indefinetly
ubuntu@ldap-client:~$ sudo nvim /etc/auto.home
* -fstype=nfs,rw,soft,intr,tcp,timeo=100,retrans=3 10.196.36.97:/NFS/home/&
# Restart autofs
ubuntu@ldap-client:~$ sudo systemctl restart autofs
```

6. Move the home directory for the **3 users/employees** created in the previous assignment to **/NFS/home**. Modify the **LDAP** server entries for the **3 users** to point to **/NFS/home** instead of **/home/**

```
# Change home directory for each user with ldapmodifyuser

ubuntu@first-instance:~$ sudo ldapmodifyuser ismail
# Enter your modifications here, end with CTRL-D.
dn: uid=ismail,ou=People,dc=dave3610,dc=com
replace: homeDirectory
homeDirectory: /NSF/home/ismail
Successfully modified user entry uid=ismail,ou=People,dc=dave3610,dc=com in
LDAP

ubuntu@first-instance:~$ sudo ldapmodifyuser amir
# Enter your modifications here, end with CTRL-D.
dn: uid=amir,ou=People,dc=dave3610,dc=com
replace: homeDirectory
homeDirectory: /NFS/home/amir
Successfully modified user entry uid=amir,ou=People,dc=dave3610,dc=com in LDAP

ubuntu@first-instance:~$ sudo ldapmodifyuser geir
# Enter your modifications here, end with CTRL-D.
dn: uid=geir,ou=People,dc=dave3610,dc=com
replace: homeDirectory
homeDirectory: /NFS/home/geir
Successfully modified user entry uid=geir,ou=People,dc=dave3610,dc=com in LDAP
```

7. The home directories for new **LDAP** users must be created on the **/NFS/home** directory
   A. Create a new user called **testuser1** to test the setup!
      - **Kerberos** and advanced authentication mechanism will not be used at this initial phase. We would in the future like to explore the possibility of integrating **NFS** with **Kerberos**, as for now, we will only use **LDAP**.

```
# Change the template for adding entries
ubuntu@first-instance:~$ sudo nvim /etc/ldapscripts/ldapscripts.conf
[…]
MTEMPLA"/NFS/home/%u
UHOMES="/NFS/home/%u"

# Create a nfsgroup
ubuntu@first-instance:~$ sudo ldapaddgroup nfsgroup 6002
Successfully added group nfsgroup to LDAP

# Create testuser1
ubuntu@first-instance:~$ sudo ldapadduser testuser1 nfsgroup
Successfully added user testuser1 to LDAP
Successfully set password for user testuser1

# Query to check that it works
ubuntu@first-instance:~$ ldapsearch -x -LLL -H ldap:/// -b dc=dave3610,dc=com
dn: dc=dave3610,dc=com
dn: uid=testuser1,ou=People,dc=dave3610,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: testuser1
sn: testuser1
givenName: testuser1
displayName: testuser1
uid: testuser1
uidNumber: 10001
gidNumber: 6002
homeDirectory: /NFS/home/testuser1
loginShell: /bin/bash
gecos: testuser1
mail: testuser1@dave3610.com

# I gave up trying to access the home directories because of the LDAP trouble
in assignment week 6
```

Assignment-9: Storage Area Network (SAN)

The goal of this week's assignment is to learn and understand the usage of iSCSI technology as a Storage Area Network solution.

Step 1: Preparing the machines needed for an iSCSI server and client

For this assignment, you will need 2 virtual machines. One of the machines will be the server(Target) and the other the client(Initiator)

You can use the machines from previous assignments..

Server (Target): 10.196.36.97

Client (Initiator): 10.196.36.113

Step 2: Setup the iSCSI server (Target)

- You will need to attach a new disk to the server. This new disk will be used for **Logical Volume Management**.
- Login to the Server.
- Create a new **Physical Volume** on the newly attached disk, then create a new **Volume Group** called **isci1** on the new disk.
- Create a logical volume named **iscsi-disk1** of size **100 MB** on volume group **iscsi1**
    - Check that the logical volume is created:
    - We will use the newly created Logical volume as block device that the iSCSI target will make available for the iSCSI initiators

```
# Create a volume on OpenStack and attatch it to the server
In OpenStack > go to volumes > +Create Volume > iscsi-disk, empty 1GB volume >
Edit volume > manage attatchments > select server instance and attatch >
attatched to /dev/sdh

# Check if the disk is available
ubuntu@first-instance:~$ sudo fdisk -l
Disk /dev/sdh: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

# Partition /dev/sdh to prepare for LVM
ubuntu@first-instance:~$ echo 'type=8e' | sudo sfdisk /dev/sdh
Checking that no-one is using this disk right now ... OK

Disk /dev/sdh: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
>>> Created a new DOS disklabel with disk identifier 0x60de4bed.
/dev/sdh1: Created a new partition 1 of type 'Linux LVM' and of size 1023 MiB.
/dev/sdh2: Done.

New situation:
Disklabel type: dos
Disk identifier: 0x60de4bed

Device     Boot Start     End Sectors  Size Id Type
/dev/sdh1        2048 2097151 2095104 1023M 8e Linux LVM

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

# Check that /dev/sdh is partitioned
ubuntu@first-instance:~$ sudo fdisk -l
Disk /dev/sdh: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x60de4bed

Device     Boot Start     End Sectors  Size Id Type
/dev/sdh1        2048 2097151 2095104 1023M 8e Linux LVM

# Create a physical volume on /dev/sdh1
ubuntu@first-instance:~$ sudo pvcreate /dev/sdh1
  Physical volume "/dev/sdh1" successfully created.

# Check that the physical volume is created
ubuntu@first-instance:~$ sudo pvdisplay -C
  PV         VG  Fmt  Attr PSize    PFree
  /dev/sdh1      lvm2 ---  1023.00m 1023.00m

# Create volume group iscsi1 on sdh1
ubuntu@first-instance:~$ sudo vgcreate iscsi1 /dev/sdh1
  Volume group "iscsi1" successfully created

# Check that iscsi1 is created
ubuntu@first-instance:~$ sudo vgdisplay -C
  VG     #PV #LV #SN Attr   VSize    VFree
  iscsi1   1   0   0 wz--n- 1020.00m 1020.00m

# Create a logical volume, iscsi-disk1, with 100MB on volume group iscsi1
ubuntu@first-instance:~$ sudo lvcreate -L 100MB -n iscsi-disk1 iscsi1
  Logical volume "iscsi-disk1" created.

# Check that the logical volume was created
ubuntu@first-instance:~$ sudo lvdisplay
  --- Logical volume ---
  LV Path                /dev/iscsi1/iscsi-disk1
  LV Name                iscsi-disk1
  VG Name                iscsi1
  LV UUID                ljWeYh-AP3R-VAqK-CcTc-sjrx-D0ei-yl9r7O
  LV Write Access        read/write
  LV Creation host, time first-instance, 2024-05-02 13:40:01 +0000
  LV Status              available
```

```
  # open                  0
  LV Size                 100.00 MiB
  Current LE              25
  Segments                1
  Allocation              inherit
  Read ahead sectors      auto
  - currently set to      256
  Block device            253:2
```

- Install the necessary iSCSI target package. In **DAVE3610**, we will use **LIO targetcli**
  - sudo apt update; sudo apt -y install targetcli-fb
- Configure **targetcli** so that:
  - Use the following IQN:
    - **iqn.2024-04.dave3610.no:Target**
  - It uses the newly created logical volume **iscsi-disk1** as block disk that can be served to iSCSI clients (Initiators)
  - sets an Access Control so that the Logical Unit Number (**LUN**) is protected by a username and a password.
    - This will make sure that a client (initiator) that wants to use the LUN must login/provide a username and a password
  - Save the configuration
- Enable the iSCSI targetcli
  - sudo systemctl enable rtslib-fb-targetctl
- Restart the service
  - sudo systemctl restart rtslib-fb-targetctl
- Check if the service is running. iSCSI target per default runs on port **3260**.
  - netstat -pnaut | grep 3260
- See the following demo on how to install and configure targetcli
  - https://asciinema.org/a/7b5qwu1e2qvny3iy0h9nbu58s

```
# Install the necessary iSCSI packages
ubuntu@first-instance:~$ sudo apt -y install targetcli-fb

# Entet targetcli environment
ubuntu@first-instance:~$ sudo targetcli
Warning: Could not load preferences file /root/.targetcli/prefs.bin.
targetcli shell version 2.1.53
Copyright 2011-2013 by Datera, Inc and others.
For help on commands, type 'help'.

/> ls
o- / ..................................................................................... [...]
  o- backstores ........................................................................... [...]
  | o- block ................................................................ [Storage Objects: 0]
  | o- fileio ............................................................... [Storage Objects: 0]
  | o- pscsi ................................................................ [Storage Objects: 0]
  | o- ramdisk .............................................................. [Storage Objects: 0]
  o- iscsi ......................................................................... [Targets: 0]
  o- loopback ...................................................................... [Targets: 0]
  o- vhost ......................................................................... [Targets: 0]
# Check which storage types targetcli supports
ubuntu@first-instance:~$ sudo targetcli /backstores ls
o- backstores ............................................................................ [...]
  o- block ................................................................ [Storage Objects: 0]
  o- fileio ............................................................... [Storage Objects: 0]
```

```
  o- pscsi ..................................................................... [Storage Objects: 0]
  o- ramdisk ................................................................... [Storage Objects: 0]
```

 # Configure targetcli to use iscsi-disk1 LVM block as block backstore

```
ubuntu@first-instance:~$ sudo targetcli /backstores/block create San-Disk-1
/dev/iscsi1/iscsi-disk1
Created block storage object San-Disk-1 using /dev/iscsi1/iscsi-disk1.
```

 # Print the config to verify

```
ubuntu@first-instance:~$ sudo targetcli /backstores ls
o- backstores ................................................................................ [...]
  o- block .................................................................... [Storage Objects: 1]
  | o- San-Disk-1 ........................... [/dev/iscsi1/iscsi-disk1 (100.0MiB) write-thru deactivated]
  |   o- alua ................................................................................ [ALUA Groups: 1]
  |     o- default_tg_pt_gp ................................................ [ALUA state: Active/optimized]
```

 # Configure iSCSI Target with specific IQN

```
ubuntu@first-instance:~$ sudo targetcli /iscsi create iqn.2024-04.dave3610.no:Target
Created target iqn.2024-04.dave3610.no:target.
Created TPG 1.
Global pref auto_add_default_portal=true
Created default portal listening on all IPs (0.0.0.0), port 3260.
```

 # Verify configuration

```
ubuntu@first-instance:~$ sudo targetcli ls
o- / ......................................................................................... [...]
  o- backstores .............................................................................. [...]
  | o- block ................................................................... [Storage Objects: 1]
  | | o- San-Disk-1 ......................... [/dev/iscsi1/iscsi-disk1 (100.0MiB) write-thru deactivated]
  | |   o- alua ................................................................ [ALUA Groups: 1]
  | |     o- default_tg_pt_gp ........................................... [ALUA state: Active/optimized]
  | o- fileio .................................................................. [Storage Objects: 0]
  | o- pscsi ................................................................... [Storage Objects: 0]
  | o- ramdisk ................................................................. [Storage Objects: 0]
  o- iscsi ........................................................................ [Targets: 1]
  | o- iqn.2024-04.dave3610.no:target ............................................... [TPGs: 1]
  |   o- tpg1 ............................................................ [no-gen-acls, no-auth]
  |     o- acls .............................................................................. [ACLs: 0]
  |     o- luns .............................................................................. [LUNs: 0]
  |     o- portals ........................................................................ [Portals: 1]
  |       o- 0.0.0.0:3260 ................................................................................ [OK]
  o- loopback ..................................................................... [Targets: 0]
  o- vhost ........................................................................ [Targets: 0]
```

 # Map backstore device to a Logical Unit Number (LUN)

```
ubuntu@first-instance:~$ sudo targetcli /iscsi/iqn.2024-04.dave3610.no:target/tpg1/luns
create /backstores/block/San-Disk-1
Created LUN 0.
```

 # Verify configuration

```
ubuntu@first-instance:~$ sudo targetcli /iscsi/iqn.2024-04.dave3610.no:target/tpg1/luns
ls
o- luns ...................................................................................... [LUNs: 1]
  o- lun0 ............................... [block/San-Disk-1 (/dev/iscsi1/iscsi-disk1) (default_tg_pt_gp)]
```

 # Create network portal to specify Target ip address (tried both target ip and localhost, but did not work)

```
ubuntu@first-instance:~$ sudo targetcli
/iscsi/iqn.2024-04.dave3610.no:target/tpg1/portals create 10.196.36.97
Using default IP port 3260
Could not create NetworkPortal in configFS
ubuntu@first-instance:~$ sudo targetcli
/iscsi/iqn.2024-04.dave3610.no:target/tpg1/portals create 192.168.10.1
Using default IP port 3260
Could not create NetworkPortal in configFS
```

 # Verify config

```
ubuntu@first-instance:~$ sudo targetcli ls
```

```
o- / ..... [...]
  o- backstores ..... [...]
  | o- block ..... [Storage Objects: 1]
  | | o- San-Disk-1 ..... [/dev/iscsi1/iscsi-disk1 (100.0MiB) write-thru activated]
  | |   o- alua ..... [ALUA Groups: 1]
  | |     o- default_tg_pt_gp ..... [ALUA state: Active/optimized]
  | o- fileio ..... [Storage Objects: 0]
  | o- pscsi ..... [Storage Objects: 0]
  | o- ramdisk ..... [Storage Objects: 0]
  o- iscsi ..... [Targets: 1]
  | o- iqn.2024-04.dave3610.no:target ..... [TPGs: 1]
  |   o- tpg1 ..... [no-gen-acls, no-auth]
  |     o- acls ..... [ACLs: 0]
  |     o- luns ..... [LUNs: 1]
  |     | o- lun0 ..... [block/San-Disk-1 (/dev/iscsi1/iscsi-disk1) (default_tg_pt_gp)]
  |     o- portals ..... [Portals: 1]
  |       o- 0.0.0.0:3260 ..... [OK]
```

 # Put Access Control on LUN

```
ubuntu@first-instance:~$ sudo targetcli /iscsi/iqn.2024-04.dave3610.no:target/tpg1/acls
create iqn.2024-04.dave3610.no:initiator
Created Node ACL for iqn.2024-04.dave3610.no:initiator
Created mapped LUN 0.
```

 # Set username and password

```
ubuntu@first-instance:~$ sudo targetcli
/iscsi/iqn.2024-04.dave3610.no:target/tpg1/acls/iqn.2024-04.dave3610.no:initiator set
auth userid=sysadmin
Parameter userid is now 'sysadmin'.
ubuntu@first-instance:~$ sudo targetcli
/iscsi/iqn.2024-04.dave3610.no:target/tpg1/acls/iqn.2024-04.dave3610.no:initiator set
auth password=sysadmin
Parameter password is now 'sysadmin'.
```

 # Verify authentication

```
ubuntu@first-instance:~$ sudo targetcli
/iscsi/iqn.2024-04.dave3610.no:target/tpg1/acls/iqn.2024-04.dave3610.no:initiator get
auth

AUTH CONFIG GROUP
=================
mutual_password=
----------------
The mutual_password auth parameter.

mutual_userid=
--------------
The mutual_userid auth parameter.

password=sysadmin
-----------------
The password auth parameter.

userid=sysadmin
---------------
The userid auth parameter.
```

 # Print out full config

```
ubuntu@first-instance:~$ sudo targetcli ls
o- / ..... [...]
  o- backstores ..... [...]
  | o- block ..... [Storage Objects: 1]
  | | o- San-Disk-1 ..... [/dev/iscsi1/iscsi-disk1 (100.0MiB) write-thru activated]
  | |   o- alua ..... [ALUA Groups: 1]
  | |     o- default_tg_pt_gp ..... [ALUA state: Active/optimized]
  | o- fileio ..... [Storage Objects: 0]
  | o- pscsi ..... [Storage Objects: 0]
  | o- ramdisk ..... [Storage Objects: 0]
  o- iscsi ..... [Targets: 1]
  | o- iqn.2024-04.dave3610.no:target ..... [TPGs: 1]
  |   o- tpg1 ..... [no-gen-acls, no-auth]
  |     o- acls ..... [ACLs: 1]
  |     | o- iqn.2024-04.dave3610.no:initiator ..... [Mapped LUNs: 1]
  |     |   o- mapped_lun0 ..... [lun0 block/San-Disk-1 (rw)]
```

```
  |     o- luns ..................................................................................... [LUNs: 1]
  |     | o- lun0 ............................................ [block/San-Disk-1 (/dev/iscsi1/iscsi-disk1) (default_tg_pt_gp)]
  |     o- portals ................................................................................ [Portals: 1]
  |       o- 0.0.0.0:3260 ......................................................................... [OK]
  o- loopback ....................................................................................... [Targets: 0]
  o- vhost .......................................................................................... [Targets: 0]

# Save config
ubuntu@first-instance:~$ sudo targetcli saveconfig
Last 10 configs saved in /etc/rtslib-fb-target/backup/.
Configuration saved to /etc/rtslib-fb-target/saveconfig.json

# Enable and start target
ubuntu@first-instance:~$ sudo systemctl enable target
Created symlink /etc/systemd/system/multi-user.target.wants/target.service →
/lib/systemd/system/target.service.

ubuntu@first-instance:~$ sudo systemctl start target

# Enable the iSCSI targetcli
ubuntu@first-instance:~$ sudo systemctl enable rtslib-fb-targetctl
Synchronizing state of rtslib-fb-targetctl.service with SysV service script with
/lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable rtslib-fb-targetctl

# Restart service
ubuntu@first-instance:~$ sudo systemctl restart rtslib-fb-targetctl

# Check if service is running
ubuntu@first-instance:~$ sudo apt install net-tools
ubuntu@first-instance:~$ netstat -pnaut | grep 3260
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp        0      0 0.0.0.0:3260            0.0.0.0:*               LISTEN      -
```

Step 3: Install and configure iSCSI client (Initiator)

- Login to the client machine
- Become root with the **sudo** command
- Install the necessary iSCSI client package. In **DAVE3610**, we will use **open-iscsi**
  - sudo apt update; sudo apt -y install open-iscsi
- Edit the file **/etc/iscsi/initiatorname.iscsi** and change its content to:
  - **iqn.2024-04.dave3610.no:initiator**
    - This is the name of the initiator that will be presented to the iSCSI target. The target will check the name against the configured ACL to grant access to the initiator.

```
# Install open-iscsi
ubuntu@ldap-client:/$ sudo apt -y install open-iscsi

# Change initiator name
ubuntu@ldap-client:/$ sudo nvim /etc/iscsi/initiatorname.iscsi
InitiatorName=iqn.2024-04.dave3610.no:initiator
```

- Most of all the configuration of **open-iscsi** is done in the file **/etc/iscsi/iscsid.conf**
- Configure **open-iscsi** so that:

- the service starts automatically when the machine starts or reboots
- it uses the **CHAP** method for authentication and set the username and password to the one configured on the server, in our case **username=sysadmin** and **password=sysadmin**

```
# Edit iscsid config by uncommenting lines
ubuntu@ldap-client:/$ sudo nvim /etc/iscsi/iscsid.conf
node.startup = automatic
node.session.auth.authmethod = CHAP
node.session.auth.username = sysadmin
node.session.auth.password = sysadmin
```

- Check that the only disks (devices) that are available for the client before it connects to the iSCSI target
  - fdisk -l | grep sd
- Enable the service and restart the service
  - systemctl enable iscsid ; systemctl restart iscsid
- Discover the target
  - iscsiadm -m discovery -t st -p The-IP_Address_of_The_Server
- Log into the target
  - iscsiadm -m node -T iqn.2024-04.dave3610.no:target -p The-IP_Address_of_The_Server -l
- Check all the disks that are now available to the client, you should see a new disk appear as **/dev/sdX** Where **X** can be **b**, **c**, **d**, **e**, **f** and so on, depending on the number of disks that you have from before.
  - fdisk -l | grep sd
- **If it appears, then congratulations! You have now a disk through iSCSI Storage Area Network**
- Reboot the machine and check again if the iSCSI disk is part of the disks the machine discovers and gets access to.
- See the following demo on how to install and configure **open-iscsi**
  - https://asciinema.org/a/6fxzlkawiwsmu280dykhoqojl

```
# Check that there are only local disks available
ubuntu@ldap-client:/$ sudo fdisk -l | grep sd
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
/dev/sda1  227328 20971486 20744159  9.9G Linux filesystem
/dev/sda14   2048    10239     8192    4M BIOS boot
/dev/sda15  10240   227327   217088  106M EFI System

# Enable and restart service
ubuntu@ldap-client:/$ sudo systemctl enable iscsid
Synchronizing state of iscsid.service with SysV service script with
/lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable iscsid
Created symlink /etc/systemd/system/sysinit.target.wants/iscsid.service →
/lib/systemd/system/iscsid.service.
ubuntu@ldap-client:/$ sudo systemctl restart iscsid

# Discover target
ubuntu@ldap-client:/$ iscsiadm -m discovery -t st -p 10.196.36.97
```

```
# Check for new disk
ubuntu@ldap-client:/$ sudo fdisk -l | grep sd
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
/dev/sda1  227328 20971486 20744159  9.9G Linux filesystem
/dev/sda14   2048    10239     8192    4M BIOS boot
/dev/sda15  10240   227327   217088  106M EFI System
# Does not work since I could not set ip on server
ubuntu@ldap-client:/$ sudo iscsiadm -m node -T iqn.2024-04.dave3610.no:initiator, -
p 10.196.36.97
iscsiadm: No records found

# If it had worked: Install lsscsi
ubuntu@ldap-client:/$ sudo apt install lsscsi

# Check for the disk
ubuntu@ldap-client:/$ sudo lsscsi # Disk should've shown up as LIO-ORG IBLOCK disk.
```

Assignment-10.1: Dynamic Host Configuration Protocol (DHCP)

The goal of this assignment is to:

- setup and configure a Dynamic Host Configuration Protocol (**DHCP**) server for a particular subnet

---

As the newly recruited system administrator, you receive a new mail from the senior system administrator asking you to carry out the following:

Hi Junior Sysadmin,

Great job in setting up the **iSCSI Target and Initiator**. We are experiencing a growth of machines and assigning IP addresses manually is no longer a viable solution.

Your task is to set up a **DHCP** server considering the following requirements:

1. Setup a DHCP server that serves clients in the domain **dave3610.com** and the subnet **192.168.10.0/24**:
2. Add an extra network interface to the Server. The Server should listen to this network interface.
3. Clients should receive:
    - IP from the range: 192.168.10.100 – 192.168.10.200
    - Domain: dave3610.com
    - Domain Name Servers: dns1.dave3610.com
4. The client **client.dave3610.com** must be given a fixed/static IP address **192.168.10.211** by the **DHCP** server
5. The default lease time for the client IP addresses must be set to **12 hours** and must not exceed **24 hours**.
6. Configure the client to get the fixed/static IP address from the **DHCP** server instead of setting it statically.

Best regards,
Senior sysadmin!

- You can't do like the Demo to ssh to the client's IP after the service is up. You have to finish step 6. first for the client to get the 192.168.10.211 IP address.
- We bump into a limitation of our current setup of the OpenStack Cloud. The 2nd network interface doesn't work as it should be. In order to do this assignement, we will do it in LXD.
    - Demo on how to prepare for an LXD virtualization platform
    - Demo on how to create, stop and delete an LXD virtual machine
    - Demo on how to attach an additional disk to an LXD virtual machine
    - Demo on how to detach/remove the additional disk from the LXD virtual machine and the LXD default storage pool

```
# Install the ISC DHCP server package
ubuntu@first-instance:~$ sudo apt install isc-dhcp-server

# Check which network interface that the machine is aware of
ubuntu@first-instance:~$ ifconfig

ens3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.196.36.97  netmask 255.255.252.0  broadcast 10.196.39.255
        inet6 fe80::f816:3eff:fe5c:be85  prefixlen 64  scopeid 0x20<link>
        ether fa:16:3e:5c:be:85  txqueuelen 1000  (Ethernet)
        RX packets 238693  bytes 26503073 (26.5 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10828  bytes 1389183 (1.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1491  bytes 146315 (146.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1491  bytes 146315 (146.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

# We see that ens3 is the main network interface used
# Change the DHCP server config to listen to the given network interface
ubuntu@first-instance:~$ sudo nvim /etc/default/isc-dhcp-server
INTERFACESv4="ens3"

# Take a backup of the dhcpd config before doing changes
ubuntu@first-instance:~$ sudo mv /etc/dhcp/dhcpd.conf
/etc/dhcp/dhcpd.conf.back

# Edit the config file
ubuntu@first-instance:~$ sudo nvim /etc/dhcp/dhcpd.conf

# Global values that are common for all subnetworks
option domain-name "dave3610.com";
option domain-name-servers dns1.dave3610.com;
default-lease-time 43200; # 12 hours in seconds
max-lease-time 86400; # 24 hours in seconds
authoritative;

# Subnetwork specification
# Here we specify the subnetwork we want the DHCP server to be responsible for
# We also specify some values such us the subnet, subnet mask and the DNS
server
subnet 192.168.10.0 netmask 255.255.255.0 {
option subnet-mask              255.255.255.0;
option domain-search            "dave3610.com";
option domain-name-servers      dns1.dave3610.com;
option routers                  192.168.10.1;
range   192.168.10.100   192.168.10.200;
}

# Here we are assigning static IP address to the host client {
hardware ethernet 08:00:27:ed:4B:8d; # Swap out after setting up client in LXD
fixed-address 192.168.10.211;
}
```

```
# Set up LXD virtualization
# Check if LXD is installed as snap (should be available in Ubuntu as default)
# Use `sudo snap install lxd` if that is not the case
ubuntu@first-instance:~$ sudo snap list
Name    Version        Rev    Tracking       Publisher   Notes
lxd     5.0.3-d921d2e  28373  5.0/stable/…   canonical✓  -

# Check LXD init options
ubuntu@first-instance:~$ sudo lxd init --help
Description:
  Configure the LXD daemon
[…]

# Initialize LXD with minimal config
ubuntu@first-instance:~$ sudo lxd init --minimal

# Check list to see if there are any containers or Vms running
ubuntu@first-instance:~$ lxc ls
To start your first container, try: lxc launch ubuntu:22.04
Or for a virtual machine: lxc launch ubuntu:22.04 --vm

+------+-------+------+------+------+-----------+
| NAME | STATE | IPV4 | IPV6 | TYPE | SNAPSHOTS |
+------+-------+------+------+------+-----------+

# Returns empty list.

# Create a network bridge in LXD
ubuntu@first-instance:~$ lxc network create lxdbr1
ipv4.address=192.168.10.1/24 ipv4.nat=false
Network lxdbr1 created

# Check the network
ubuntu@first-instance:~$ lxc network list
+--------+----------+---------+----------------+--------------------------+-------------+---------+---------+
|  NAME  |   TYPE   | MANAGED |      IPV4      |          IPV6            | DESCRIPTION | USED BY |  STATE  |
+--------+----------+---------+----------------+--------------------------+-------------+---------+---------+
| ens3   | physical | NO      |                |                          |             | 0       |         |
+--------+----------+---------+----------------+--------------------------+-------------+---------+---------+
| lxdbr0 | bridge   | YES     | 10.78.132.1/24 | fd42:e8d9:eb6a:9c8d::1/64 |             | 2       | CREATED |
+--------+----------+---------+----------------+--------------------------+-------------+---------+---------+
| lxdbr1 | bridge   | YES     | 192.168.10.1/24 | fd42:ddd1:8f81:62f8::1/64 |             | 0       | CREATED |
+--------+----------+---------+----------------+--------------------------+-------------+---------+---------+

# Change isc-dhcp-server file to use the newly created bridge
ubuntu@first-instance:~$ sudo nvim /etc/default/isc-dhcp-server
INTERFACESv4="lxdbr1"

# Launch an Ubuntu container to use as client with the network bridge
ubuntu@first-instance:~$ lxc launch ubuntu:20.04 client-dhcp --profile default
--network lxdbr1
Creating client-dhcp
Starting client-dhcp

# Obtain the MAC address for the client container
ubuntu@first-instance:~$ lxc config show client-dhcp --expanded
architecture: x86_64
config:
[…]
  volatile.eth0.hwaddr:  00:16:3e:d2:b4:6d
[…]
```

```
# Change the MAC address on the server dhcp config
ubuntu@first-instance:~$ sudo nvim /etc/dhcp/dhcpd.conf

host client {
hardware ethernet 00:16:3e:f8:05:94;
fixed-address 192.168.10.211;
}

# Restart the DHCP service
ubuntu@first-instance:~$ sudo systemctl restart isc-dhcp-server

# Check if the service is running
ubuntu@first-instance:~$ sudo systemctl status isc-dhcp-server
Active: active (running) since Fri 2024-05-03 12:11:31 UTC; 10s ago

# Restart client container
ubuntu@first-instance:~$ lxc restart client-dhcp

# Check the client IP address
ubuntu@first-instance:~$ lxc list
+-------------+---------+----------------------+-----------------------------------------------+-----------+-----------+
|    NAME     |  STATE  |         IPV4         |                    IPV6                       |   TYPE    | SNAPSHOTS |
+-------------+---------+----------------------+-----------------------------------------------+-----------+-----------+
| client-dhcp | RUNNING | 192.168.10.211 (eth0) | fd42:ddd1:8f81:62f8:216:3eff:fed2:b46d (eth0) | CONTAINER | 0         |
+-------------+---------+----------------------+-----------------------------------------------+-----------+-----------+

# The client now runs on 192.168.10.211
```

Assignment-10.2: Domain Name Server (DNS)

The goal of this assignment is to:

- setup and configure a primary Domain Name Server (**DNS**)

As the newly recruited system administrator, you receive a new mail from the senior system administrator asking you to carry out the following:

Hi Junior Sysadmin,
Great job in setting up the **DHCP** server. We are experiencing a growth of machines and assigning domain names manually is no longer a viable solution. Using **/etc/hosts** for domain names is no longer acceptable either and therefore we need a Domain Name Service (**DNS**) that will serve our clients/machines locally and eventually work as the authoritative **DNS** server for our domain **dave3610.com**.

Your task is to set up a **DNS** server, considering the following requirements:

1. The **DNS** server should be installed on the same machine as the **LDAP**, **NFS** or **iSCSI** server.
2. The **DNS** server should have the same IPv4 as the **LDAP**, **NFS** or **iSCSI** server, namely **192.168.10.1**
3. The **DNS** server should act as a **Caching Nameserver** and use the IP **8.8.8.8** and **8.8.4.4** as the forwarders
4. The **DNS** server should be responsible for the domain **dave3610.com** and be able to respond to queries regarding:
    - Forward-lookups:
        - Should return the IPv4 address when a client queries for the fully qualified domain name (FQDN) www.dave3610.com.
            - Example: nslookup www.dave3610.com 192.168.10.1
    - Reverse-lookups:
        - Should return the fully qualified domain name (FQDN) www.dave3610.com when a client queries for the IPv4 address.
            - Example: nslookup 192.168.10.1
5. Configure an IPv4 record for **www** that points to **192.168.10.10**
6. Configure an IPv4 record for **client** that points to **192.168.10.211**
7. The **DNS** server must be configured with **Canonical Names** that point to the IPv4 address of the **www** server for the following:
    - **webmail** and **mail**
8. Configure a reverse-lookups for the following:
    - 192.168.10.1, 192.168.10.10 and 192.168.10.211

Best regards,
Senior sysadmin!

(Tips: If you set the bind's cache directory to be the exact location you see in the Demo, you might need to correct the permissions of the cache directory too. Otherwise, the bind service will not be started.)

```
# Install the necessary packages
ubuntu@first-instance:~$ sudo apt -y install bind9 bind9-host bind9utils
dnsutils




# Configure named.conf.options to tell BIND to forward other lookups
ubuntu@first-instance:~$ sudo nvim /etc/bind/named.conf.options

options {
        directory "/var/cache/bind";

        forwarders {
                8.8.8.8;
                8.8.4.4;
        };


        dnssec-validation no;
        //listen-on-v6 { any; };

        listen-on { 192.168.10.1; };
        allow-query { any; };
        recursion yes;
};





# Configure zones for dave3610.com
ubuntu@first-instance:~$ sudo nvim /etc/bind/named.conf.local

zone "dave3610.com" {
    type master;
    file "/etc/bind/db.dave3610.com";
};

zone "10.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10.168.192";
};





# Create forward zone file
ubuntu@first-instance:~$ sudo cp /etc/bind/db.local /etc/bind/db.dave3610.com
ubuntu@first-instance:~$ sudo nvim /etc/bind/db.dave3610.com
```

```
;
; BIND data file for dave3610.com
;
$TTL    604800
@       IN      SOA     dave3610.com. root.dave3610.com. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      dave3610.com.
@       IN      A       192.168.10.1
www     IN      A       192.168.10.10
client  IN      A       192.168.10.211
webmail IN      CNAME   www
mail    IN      CNAME   www
```

 # Create reverse zone file and configure PTR records

```
ubuntu@first-instance:~$ sudo cp /etc/bind/db.127 /etc/bind/db.10.168.192
ubuntu@first-instance:~$ sudo nvim /etc/bind/db.10.168.192

;
; BIND reverse data file for local loopback interface
;
$TTL    604800
@       IN      SOA     dave3610.com. root.dave3610.com. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      dave3610.com.
1       IN      PTR     dave3610.com.
10      IN      PTR     www.dave3610.com.
211     IN      PTR     client.dave3610.com.
```

 # Restart BIND9 and test configuration

```
ubuntu@first-instance:~$ sudo systemctl restart bind9
ubuntu@first-instance:~$ sudo systemctl status bind9
Active: active (running) since Fri 2024-05-03 13:43:54 UTC; 6s ago
```

 # Test DNS configuration

```
ubuntu@first-instance:~$ nslookup www.dave3610.com 192.168.10.1
Server:           192.168.10.1
Address:    192.168.10.1#53

Name: www.dave3610.com
Address: 192.168.10.10

@first-instance:~$ nslookup 192.168.10.10
** server can't find 10.10.168.192.in-addr.arpa: NXDOMAIN
```

 # For some reason, nslookup can’t manage to do reverse lookup
 # Dig for some reason can. I won’t bother finding out why

```
ubuntu@first-instance:~$ dig -x 192.168.10.10 @192.168.10.1

; <<>> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <<>> -x 192.168.10.10 @192.168.10.1
;; global options: +cmd
```

```
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13348
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: ebc2d4a0fb80e13d010000006634f0fd77626900f10c75a7 (good)
;; QUESTION SECTION:
;10.10.168.192.in-addr.arpa.  IN    PTR

;; ANSWER SECTION:
10.10.168.192.in-addr.arpa. 604800 IN    PTR   www.dave3610.com.

;; Query time: 0 msec
;; SERVER: 192.168.10.1#53(192.168.10.1) (UDP)
;; WHEN: Fri May 03 14:13:17 UTC 2024
;; MSG SIZE  rcvd: 113


ubuntu@first-instance:~$ dig @192.168.10.1 www.dave3610.com

; <<>> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <<>> @192.168.10.1 www.dave3610.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 38314
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 6a86aa1699e9764d010000006634f1173e9844b2022a4d3f (good)
;; QUESTION SECTION:
;www.dave3610.com.            IN    A

;; ANSWER SECTION:
www.dave3610.com. 604800      IN    A     192.168.10.10

;; Query time: 0 msec
;; SERVER: 192.168.10.1#53(192.168.10.1) (UDP)
;; WHEN: Fri May 03 14:13:43 UTC 2024
;; MSG SIZE  rcvd: 89
```

Assignment-11: Email system

The **DAVE3610** University has for many years used an external email provider for its email systems. After an increased number of students and staff in the past 3 years, security and privacy concerns with the current email solution, the University decided to operate its own email system. The University policy is to only use open standards and avoid proprietary solutions. The IT department has been given a task to build a solution based on Open Source technologies with the following specifications. The system must implement the specifications given.

Hi Junior Sysadmin,
Great job so far in this semester. We want you as a final task to set up an **Email system** based on the requirements specified above.

A good place to start for resources is:

- https://ubuntu.com/server/docs/mail-introduction

Good luck with this final assignment!

Best regards,
Senior Sysadmin!

MTA/SMTP

- A scalable **SMTP** server:
  - **Postfix** is the required **SMTP** server for the University.
  - **NB!** You are not allowed to use packages such as **mail-stack-delivery** or similar. You must install postfix package and configure it from scratch!

```
# Install the Postfix packagefor SMTP/MTA server
ubuntu@first-instance:~$ sudo DEBIAN_FRONTEND=noninteractive apt install -y
postfix
# Use DEBIAN_FRONTEND=noninteractive to avoid pre-configured parameters

# Check the content of the two main config files for Postfix
ubuntu@first-instance:~$ sudo cat /etc/postfix/main.cf
ubuntu@first-instance:~$ sudo cat /etc/postfix/master.cf

# Check the main configs wih the postconf tool
ubuntu@first-instance:~$ sudo postconf -n
ubuntu@first-instance:~$ sudo postconf

# Stop Postfix before configuring the MTA server
ubuntu@first-instance:~$ sudo /etc/init.d/postfix stop
Stopping postfix (via systemctl): postfix.service.

# Make a backup of the config files before editing configs
ubuntu@first-instance:~$ sudo cp /etc/postfix/main.cf /etc/postfix/main.cf.back
ubuntu@first-instance:~$ sudo bash -c 'cat /dev/null > /etc/postfix/main.cf'
# Empty the config file

# Add the Fully Qualified Domain Name of the server to the file /etc/mailname
ubuntu@first-instance:~$ sudo bash -c 'echo server.dave3610.com >
/etc/mailname'

# Add entries to main.cf with the postconf tool (could have been done manually
too)
# Specify domain name
ubuntu@first-instance:~$ sudo bash -c 'echo server.dave3610.com >
/etc/mailname'

# Specify host name
ubuntu@first-instance:~$ sudo postconf -e 'mydomain = dave3610.com'

# Specify host name
ubuntu@first-instance:~$ sudo postconf -e 'myhostname = server.dave3610.com'

# Specify domain name that locally posted mail appears to come from and is
delivered to
ubuntu@first-instance:~$ sudo postconf -e 'myorigin = /etc/mailname'

# Specify format in which received mail will be stored. We use Maildir
ubuntu@first-instance:~$ sudo postconf -e 'home_mailbox = Maildir/'

# Specify which domain the server is responsible and will receive mail for
(Shared Domain with System Accounts)
ubuntu@first-instance:~$ sudo postconf -e 'mydestination = dave3610.com,
server.dave3610.com, server, localhost.localdomain, localhost'

# Specify which machine or network that can send mail through the SMTP/MTA
server (used when SMTP-AUTH is not configured)
```

```
ubuntu@first-instance:~$ sudo postconf -e 'mynetworks = 127.0.0.0/8
[::ffff:127.0.0.0]/104 [::1]/128 192.168.10.0/24'

# Specify which network interface/card address to use
ubuntu@first-instance:~$ sudo postconf -e 'inet_interfaces = all'

# Specify which protocol to use. We'll use only Ipv4
ubuntu@first-instance:~$ sudo postconf -e 'inet_protocols = ipv4'

# Specify max size of any individual mailbox or maildir file
ubuntu@first-instance:~$ sudo postconf -e 'mailbox_size_limit = 0' # no limit

# Specify how to seperate a user name from its extension
ubuntu@first-instance:~$ sudo postconf -e 'recipient_delimiter = +'

# Specify the file containing aliases which provides system-wide mechanims to
redirect mail for local recipients
ubuntu@first-instance:~$ sudo postconf -e 'alias_database = hash:/etc/aliases'
ubuntu@first-instance:~$ sudo postconf -e 'alias_maps = hash:/etc/aliases'

# Check configuration
ubuntu@first-instance:~$ sudo cat /etc/postfix/main.cf

mydomain = dave3610.com
myhostname = server.dave3610.com
myorigin = /etc/mailname
home_mailbox = Maildir/
mydestination = dave3610.com, server.dave3610.com, server,
localhost.localdomain, localhost
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 192.168.10.0/24
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
recipient_delimiter = +
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases

# Use postconf to check the configuration
ubuntu@first-instance:~$ sudo postconf -n
# Its the same

# Restart server and check the logs
ubuntu@first-instance:~$ sudo /etc/init.d/postfix restart; sudo tail -40
/var/log/mail.log

Restarting postfix (via systemctl): postfix.service.
May  6 10:03:56 first-instance postfix/postfix-script[24597]: starting the Postfix mail system
May  6 10:03:56 first-instance postfix/master[24599]: daemon started -- version 3.6.4,
configuration /etc/postfix
May  6 10:09:38 first-instance postfix/postfix-script[24719]: stopping the Postfix mail system
May  6 10:09:38 first-instance postfix/master[24599]: terminating on signal 15
May  6 10:36:00 first-instance postfix[24876]: Postfix is running with backwards-compatible default
settings
May  6 10:36:00 first-instance postfix[24876]: See http://www.postfix.org/COMPATIBILITY_README.html
for details
May  6 10:36:00 first-instance postfix[24876]: To disable backwards compatibility use "postconf
compatibility_level=3.6" and "postfix reload"
May  6 10:36:00 first-instance postfix/postfix-script[25105]: starting the Postfix mail system
May  6 10:36:00 first-instance postfix/master[25107]: daemon started -- version 3.6.4,
configuration /etc/postfix

# Install swak to send email in cli
ubuntu@first-instance:~$ sudo DEBIAN_FRONTEND=noninteractive apt install -y
swaks
```

```
# Send email from user user ubuntu to user ubuntu
ubuntu@first-instance:~$ swaks -tlso --to ubuntu@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: Testing
MTA by Sysadmin'  --body 'Testing a basic SMTP/MTA functions'

=== Trying 127.0.0.1:25...
=== Connected to 127.0.0.1.
<** Timeout (30 secs) waiting for server response
 -> QUIT

# It fails. Learned that I needed to add smtpd relay restrictions
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_relay_restrictions =
permit_mynetworks permit_sasl_authenticated defer_unauth_destination'
ubuntu@first-instance:~$ sudo postconf -n
[…]
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated
defer_unauth_destination

# Restart server and check logs again
ubuntu@first-instance:~$ sudo /etc/init.d/postfix restart

# Try again to send email from user user ubuntu to user ubuntu
ubuntu@first-instance:~$ swaks -tlso --to ubuntu@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: Testing
MTA by Sysadmin'  --body 'Testing a basic SMTP/MTA functions'

=== Trying 127.0.0.1:25...
=== Connected to 127.0.0.1.
<-  220 server.dave3610.com ESMTP Postfix
 -> EHLO first-instance
<-  250-server.dave3610.com
<-  250-PIPELINING
<-  250-SIZE 10240000
<-  250-VRFY
<-  250-ETRN
<-  250-ENHANCEDSTATUSCODES
<-  250-8BITMIME
<-  250-DSN
<-  250 CHUNKING
*** Host did not advertise STARTTLS
 -> MAIL FROM:<ubuntu@dave3610.com>
<-  250 2.1.0 Ok
 -> RCPT TO:<ubuntu@dave3610.com>
<-  250 2.1.5 Ok
 -> DATA
<-  354 End data with <CR><LF>.<CR><LF>
 -> Date: Mon, 06 May 2024 10:48:13 +0000
 -> To: ubuntu@dave3610.com
 -> From: ubuntu@dave3610.com
 -> Subject: test Mon, 06 May 2024 10:48:13 +0000
 -> Message-Id: <20240506104813.025714@first-instance>
 -> X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
 -> X-Custom-Header: Testing MTA by Sysadmin
 ->
 -> Testing a basic SMTP/MTA functions
 ->
 ->
 -> .
<-  250 2.0.0 Ok: queued as 4D44EAAC4
 -> QUIT
<-  221 2.0.0 Bye
=== Connection closed with remote host.

# Check logs to see if mail is delivered by postfix
ubuntu@first-instance:~$ sudo grep 'delivered to maildir' /var/log/mail.log
May  6 10:48:13 first-instance postfix/local[25719]: 4D44EAAC4:
to=<ubuntu@dave3610.com>, relay=local, delay=0.22, delays=0.13/0.01/0/0.08,
```

```
dsn=2.0.0, status=sent (delivered to maildir)

# Check ubuntu MailDir directory to see mail and read the content
ubuntu@first-instance:~$ sudo bash -c 'cat /home/ubuntu/Maildir/new/*'

Return-Path: <ubuntu@dave3610.com>
X-Original-To: ubuntu@dave3610.com
Delivered-To: ubuntu@dave3610.com
Received: from first-instance (localhost [127.0.0.1])
      by server.dave3610.com (Postfix) with ESMTP id 4D44EAAC4
      for <ubuntu@dave3610.com>; Mon,  6 May 2024 10:48:13 +0000 (UTC)
Date: Mon, 06 May 2024 10:48:13 +0000
To: ubuntu@dave3610.com
From: ubuntu@dave3610.com
Subject: test Mon, 06 May 2024 10:48:13 +0000
Message-Id: <20240506104813.025714@first-instance>
X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
X-Custom-Header: Testing MTA by Sysadmin

Testing a basic SMTP/MTA functions

# Test if we can send email with TLS/SSL with swaks
ubuntu@first-instance:~$ swaks -tls --to ubuntu@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: Testing
MTA by Sysadmin'  --body 'Testing if Postfix supports SSL/TLS'

=== Trying 127.0.0.1:25...
=== Connected to 127.0.0.1.
<-  220 server.dave3610.com ESMTP Postfix
 -> EHLO first-instance
<-  250-server.dave3610.com
<-  250-PIPELINING
<-  250-SIZE 10240000
<-  250-VRFY
<-  250-ETRN
<-  250-ENHANCEDSTATUSCODES
<-  250-8BITMIME
<-  250-DSN
<-  250 CHUNKING
*** Host did not advertise STARTTLS
 -> QUIT
<-  221 2.0.0 Bye
=== Connection closed with remote host.

# Under installation of Postfix it was installed a self-signed SSL/TLS
certificate. However, we will create our own self-signed cert with openssl
ubuntu@first-instance:~$ sudo openssl req -x509 -nodes -days 365 -newkey
rsa:2048 -keyout /etc/ssl/private/postfix.key -out /etc/ssl/certs/postfix.crt

Country Name: NO
State/Province Name: Oslo
Locality Name: Oslo
Organization Name: dave3610
Organizational Unit Name: dave3610
Common Name: server.dave3610.com
Email Address:

# Configure Postfix to use the self-signed SSL/TLS cert
# Add comment to main.cf telling where the TLS config starts
ubuntu@first-instance:~$ sudo bash -c 'echo  >> /etc/postfix/main.cf'
ubuntu@first-instance:~$ sudo bash -c 'echo ##--------------TLS-
parameters--------------------------- >> /etc/postfix/main.cf'
```

```
# Configure TLS with the postconf command
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_tls_cert_file =
/etc/ssl/certs/postfix.crt'
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_tls_key_file =
/etc/ssl/private/postfix.key'
ubuntu@first-instance:~$ sudo postconf -e  smtpd_use_tls=yes

# Check config with the postconf command
ubuntu@first-instance:~$ sudo postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
mydestination = dave3610.com, server.dave3610.com, server,
localhost.localdomain, localhost
mydomain = dave3610.com
myhostname = server.dave3610.com
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 192.168.10.0/24
myorigin = /etc/mailname
recipient_delimiter = +
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated
defer_unauth_destination
smtpd_tls_cert_file = /etc/ssl/certs/postfix.crt
smtpd_tls_key_file = /etc/ssl/private/postfix.key
smtpd_use_tls = yes

# Reload Postfix so it uses the new config
ubuntu@first-instance:~$ sudo postfix reload
postfix/postfix-script: refreshing the Postfix mail system

# Try to send email with TLS/SSL with swaks again
ubuntu@first-instance:~$ swaks -tls --to ubuntu@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: Testing
MTA by Sysadmin'  --body 'Testing if Postfix supports SSL/TLS'

=== Trying 127.0.0.1:25...
=== Connected to 127.0.0.1.
<-  220 server.dave3610.com ESMTP Postfix
 -> EHLO first-instance
<-  250-server.dave3610.com
<-  250-PIPELINING
<-  250-SIZE 10240000
<-  250-VRFY
<-  250-ETRN
<-  250-STARTTLS
<-  250-ENHANCEDSTATUSCODES
<-  250-8BITMIME
<-  250-DSN
<-  250 CHUNKING
 -> STARTTLS
<-  220 2.0.0 Ready to start TLS
=== TLS started with cipher TLSv1.3:TLS_AES_256_GCM_SHA384:256
=== TLS no local certificate set
=== TLS peer DN="/C=NO/ST=Oslo/L=Oslo/O=dave3610/OU=dave3610/CN=server.dave3610.com"
 ~> EHLO first-instance
<~  250-server.dave3610.com
<~  250-PIPELINING
<~  250-SIZE 10240000
<~  250-VRFY
<~  250-ETRN
<~  250-ENHANCEDSTATUSCODES
<~  250-8BITMIME
<~  250-DSN
<~  250 CHUNKING
 ~> MAIL FROM:<ubuntu@dave3610.com>
<~  250 2.1.0 Ok
 ~> RCPT TO:<ubuntu@dave3610.com>
<~  250 2.1.5 Ok
 ~> DATA
```

```
<~  354 End data with <CR><LF>.<CR><LF>
 ~> Date: Mon, 06 May 2024 11:07:22 +0000
 ~> To: ubuntu@dave3610.com
 ~> From: ubuntu@dave3610.com
 ~> Subject: test Mon, 06 May 2024 11:07:22 +0000
 ~> Message-Id: <20240506110722.025777@first-instance>
 ~> X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
 ~> X-Custom-Header: Testing MTA by Sysadmin
 ~>
 ~> Testing if Postfix supports SSL/TLS
 ~>
 ~>
 ~> .
<~  250 2.0.0 Ok: queued as 6445CAAC8
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.
```

 # Check logs to see if mail is delivered

```
ubuntu@first-instance:~$ sudo grep 'delivered' /var/log/mail.log
May  6 11:07:23 first-instance postfix/local[25783]: 6445CAAC8:
to=<ubuntu@dave3610.com>, relay=local, delay=0.76, delays=0.3/0.02/0/0.44,
dsn=2.0.0, status=sent (delivered to maildir)
```

 # Read the mail stored under Maildir

```
ubuntu@first-instance:~$ sudo bash -c 'cat /home/ubuntu/Maildir/new/*'

Return-Path: <ubuntu@dave3610.com>
X-Original-To: ubuntu@dave3610.com
Delivered-To: ubuntu@dave3610.com
Received: from first-instance (localhost [127.0.0.1])
      by server.dave3610.com (Postfix) with ESMTPS id 6445CAAC8
      for <ubuntu@dave3610.com>; Mon,  6 May 2024 11:07:22 +0000 (UTC)
Date: Mon, 06 May 2024 11:07:22 +0000
To: ubuntu@dave3610.com
From: ubuntu@dave3610.com
Subject: test Mon, 06 May 2024 11:07:22 +0000
Message-Id: <20240506110722.025777@first-instance>
X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
X-Custom-Header: Testing MTA by Sysadmin

Testing if Postfix supports SSL/TLS
```

IMAP with SSL/TLS

- A secure (**SSL/TLS** protected) **IMAP** server that will allow the employees to retrieve their email through a mail client:
  - **Dovecot** is the required **IMAP** server for the University.
  - **NB!** You must use a self-signed SSL/TLS certificate generated by you and not Dovecot
  - **Thunderbird** is the desired mail client for the University.

```
# Install the Dovecot package for our IMAP server (noninteractive to not get
pre-configurations)
ubuntu@first-instance:~$ sudo DEBIAN_FRONTEND=noninteractive apt install -qq -y
dovecot-common dovecot-imapd

# Check the main config file for dovecot
ubuntu@first-instance:~$ sudo cat /etc/dovecot/dovecot.conf

# Check the config of dovecot with the dovecot command
ubuntu@first-instance:~$ sudo dovecot -n
# The same config, but more readable

# List all the config files for dovecot
ubuntu@first-instance:~$ ls -l /etc/dovecot/conf.d/
-rw-r--r-- 1 root root  5248 Aug  6  2021 10-auth.conf
[…]
-rw-r--r-- 1 root root  2182 Aug  6  2021 auth-system.conf.ext

# Stop the Dovecot service before making config changes
ubuntu@first-instance:~$ sudo service dovecot stop

# Take a backup of the original config file and empty the old one
ubuntu@first-instance:~$ sudo cp /etc/dovecot/dovecot.conf
/etc/dovecot/dovecot.conf.back
ubuntu@first-instance:~$ sudo bash -c 'cat /dev/null >
/etc/dovecot/dovecot.conf'

# Create a config file based on the config from the teacher
ubuntu@first-instance:~$ nvim firstdovecot.conf

protocols = imap
mail_location = maildir:~/Maildir
listen = *
disable_plaintext_auth = no
auth_mechanisms = plain login
service auth {

      unix_listener /var/spool/postfix/private/auth {

            mode = 0666

      }
}
!include conf.d/*.conf

!include_try local.conf

# Replace the dovecot.conf with this config file
ubuntu@first-instance:~$ sudo cp firstdovecot.conf /etc/dovecot/dovecot.conf
```

```
# Comment out mail_location in 10-mail.conf since it is already configured in
the main config

# First find out which line to comment out
ubuntu@first-instance:~$ grep -n mail_location /etc/dovecot/conf.d/10-mail.conf
13:# path given in the mail_location setting.
24:#   mail_location = maildir:~/Maildir
25:#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
26:#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
30:mail_location = mbox:~/mail:INBOX=/var/mail/%u
56:  # mail_location, which is also the default for it.

# Use sed to comment out line 30
ubuntu@first-instance:~$ sudo sed -i '30 s/^/#/' /etc/dovecot/conf.d/10-
mail.conf

# Check that it worked
ubuntu@first-instance:~$ grep -n mail_location /etc/dovecot/conf.d/10-mail.conf
30:#mail_location = mbox:~/mail:INBOX=/var/mail/%u

# Start the Dovecot server
ubuntu@first-instance:~$ sudo service dovecot start

# Install getMail to test our IMAP server from cli
ubuntu@first-instance:~$ sudo DEBIAN_FRONTEND=noninteractive apt install -qq -y
getmail6

# Configure getMail and create some directories for where it will store
retrieved mail from the IMAP server
ubuntu@first-instance:~$ sudo mkdir -p /home/sysadmin/.getmail/{cur,new,tmp}
2>/dev/null 1>/dev/null
ubuntu@first-instance:~$ sudo chown -R sysadmin: /home/sysadmin/.getmail/

# Create config files based on the teachers config
ubuntu@first-instance:~$ nvim getmail-IMAP-sysadmin-without-SSL

[retriever]
type = SimpleIMAPRetriever
server = 127.0.0.1
port = 143
username = sysadmin
password = pagh3377

[destination]
type = Maildir
path = /home/sysadmin/.getmail/
user = sysadmin

ubuntu@first-instance:~$ nvim getmail-IMAP-sysadmin-with-SSL

[retriever]
type = SimpleIMAPSSLRetriever
server = 127.0.0.1
username = sysadmin
password = pagh3377

[destination]
type = Maildir
path = /home/sysadmin/.getmail/mail/
user = sysadmin
```

```
# Move config files to .getmail directory
ubuntu@first-instance:~$ sudo cp getmail-IMAP-sysadmin*
/home/sysadmin/.getmail/
ubuntu@first-instance:~$ sudo chown -R sysadmin: /home/sysadmin/.getmail/

# Try to send mail with swaks and retrieve it with getmail
ubuntu@first-instance:~$ swaks -tlso --to sysadmin@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: Testing
IMAP'  --body 'Testing IMAP with Normal Mail'

=== Trying 127.0.0.1:25...
=== Connected to 127.0.0.1.
<-  220 server.dave3610.com ESMTP Postfix
 -> EHLO first-instance
<-  250-server.dave3610.com
<-  250-PIPELINING
<-  250-SIZE 10240000
<-  250-VRFY
<-  250-ETRN
<-  250-STARTTLS
<-  250-ENHANCEDSTATUSCODES
<-  250-8BITMIME
<-  250-DSN
<-  250 CHUNKING
 -> STARTTLS
<-  220 2.0.0 Ready to start TLS
=== TLS started with cipher TLSv1.3:TLS_AES_256_GCM_SHA384:256
=== TLS no local certificate set
=== TLS peer DN="/C=NO/ST=Oslo/L=Oslo/O=dave3610/OU=dave3610/CN=server.dave3610.com"
 ~> EHLO first-instance
<~  250-server.dave3610.com
<~  250-PIPELINING
<~  250-SIZE 10240000
<~  250-VRFY
<~  250-ETRN
<~  250-ENHANCEDSTATUSCODES
<~  250-8BITMIME
<~  250-DSN
<~  250 CHUNKING
 ~> MAIL FROM:<ubuntu@dave3610.com>
<~  250 2.1.0 Ok
 ~> RCPT TO:<sysadmin@dave3610.com>
<~  250 2.1.5 Ok
 ~> DATA
<~  354 End data with <CR><LF>.<CR><LF>
 ~> Date: Mon, 06 May 2024 11:45:40 +0000
 ~> To: sysadmin@dave3610.com
 ~> From: ubuntu@dave3610.com
 ~> Subject: test Mon, 06 May 2024 11:45:40 +0000
 ~> Message-Id: <20240506114540.029328@first-instance>
 ~> X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
 ~> X-Custom-Header: Testing IMAP
 ~>
 ~> Testing IMAP with Normal Mail
 ~>
 ~>
 ~> .
<~  250 2.0.0 Ok: queued as 0D5C2B03B
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.

# Check if mail is delivered
ubuntu@first-instance:~$ sudo grep 'delivered to maildir' /var/log/mail.log
May  6 11:45:42 first-instance postfix/local[29333]: 1E97EB040:
to=<ubuntu@dave3610.com>, relay=local, delay=0.72, delays=0.27/0/0/0.44,
dsn=2.0.0, status=sent (delivered to maildir)

# Read the content in sysadmins MailDir
ubuntu@first-instance:~$ sudo bash -c 'cat /home/sysadmin/Maildir/new/*'

Return-Path: <ubuntu@dave3610.com>
X-Original-To: sysadmin@dave3610.com
Delivered-To: sysadmin@dave3610.com
Received: from first-instance (localhost [127.0.0.1])
```

```
      by server.dave3610.com (Postfix) with ESMTPS id 7943AB040
      for <sysadmin@dave3610.com>; Mon,  6 May 2024 11:56:53 +0000 (UTC)
Date: Mon, 06 May 2024 11:56:53 +0000
To: sysadmin@dave3610.com
From: ubuntu@dave3610.com
Subject: test Mon, 06 May 2024 11:56:53 +0000
Message-Id: <20240506115653.029411@first-instance>
X-Mailer: swaks v20201014.0 jetmore.org/john/code/swaks/
X-Custom-Header: Testing IMAP

Testing IMAP with Normal Mail
# Check if there is any mail stored in .getmail/new
ubuntu@first-instance:~$ ls -la  /home/sysadmin/.getmail/new/
# Nothing there

# Retrieve mail with getmail
ubuntu@first-instance:~$ sudo getmail  --rcfile
/home/sysadmin/.getmail/getmail-IMAP-sysadmin-without-SSL --getmaildir
/home/sysadmin
getmail version 6.18.4
Copyright (C) 1998-2021 Charles Cazabon and others. Licensed under GNU GPL
version 2.
SimpleIMAPRetriever:sysadmin@127.0.0.1:143: read_all and not delete -- all
messages will be retrieved each time getmail is run
SimpleIMAPRetriever:sysadmin@127.0.0.1:143:
  msg 1/2 (629 bytes) delivered
  msg 2/2 (613 bytes) delivered
  2 messages (1242 bytes) retrieved, 0 skipped

# Check files retrieved by getmail
ubuntu@first-instance:~$ ls -la  /home/sysadmin/.getmail/new/
-rw------- 1 sysadmin sysadmin  802 May  6 12:03 1714997028.M846668P29442Q0.first-instance
-rw------- 1 sysadmin sysadmin  786 May  6 12:03 1714997029.M359008P29443Q1.first-instance

# Check content of file
ubuntu@first-instance:~$ sudo bash -c 'cat /home/sysadmin/.getmail/new/*'
# Mails get printed out

# Open ports in OpenStack
Go to security groups > Add rule for port 143 (IMAP) and 25 (SMTP)

# Add new account in Thunderbird
Add new account > sysadmin@dave3610.com, password > Try to automatically find
with DNS lookup, but fail > Enter server IP and port > Ignore warnings > Add
```

Webmail

- A secure (**SSL/TLS** protected) web-based solution that will allow the employees and students to retrieve their email through the web (**Webmail**):
  - **SquirrelMail** is the required webmail software for the University.
  - **NB!** You must use a self-signed SSL/TLS certificate generated by you and not Apache
  - **NB!** Squirrelmail stops providing DEB packages to Ubuntu since 16.10. You have to install it from source

```
# Install Squirrelmail from source
https://squirrelmail.org/download.php
Use dev version 1.5.2 – Stable uses php4
ubuntu@first-instance:~$ wget https://squirrelmail.org/countdl.php?fileurl=http
%3A%2F%2Fsnapshots.squirrelmail.org%2Fsquirrelmail-20240506_0200-
SVN.devel.tar.gz

# The filename is fucked, so rename it
ubuntu@first-instance:~$ mv regergergergrewfsquirrelmail-20240506_0200-
SVN.devel.tar.gz squirrelmail-20240506_0200-SVN.devel.tar.gz

# Unpack tarball
ubuntu@first-instance:~$ tar -xvzf squirrelmail-20240506_0200-SVN.devel.tar.gz

# Read the documentation from squirrelmail
ubuntu@first-instance:~$ cat squirrelmail.devel/doc/INSTALL

# Move the whole squirrelmail.devel directory to /var/www/html
ubuntu@first-instance:~$ sudo mv squirrelmail.devel /var/www/html/squirrelmail
ubuntu@first-instance:~$ ls /var/www/html
index.nginx-debian.html  squirrelmail.devel

# Install apache
ubuntu@first-instance:~$ sudo apt install apache2

# Install mysql-server
ubuntu@first-instance:~$ sudo apt install mysql-server

# Install PHP modules
ubuntu@first-instance:~$ sudo apt install php libapache2-mod-php php-mysql

# Restart apache and check status
ubuntu@first-instance:~$ sudo systemctl restart apache2
ubuntu@first-instance:~$ sudo systemctl status apache2
Active: active (running) since Mon 2024-05-06 13:13:38 UTC; 6s ago

# Open port 80 in OpenStack
Security groups > Default > manage rules > Add rule > HTTP > Add

# Generate SSL cert
ubuntu@first-instance:~$ sudo openssl req -x509 -nodes -days 365 -newkey
rsa:2048 -keyout /etc/ssl/private/apache-selfsigned.key -out
/etc/ssl/certs/apache-selfsigned.crt

Country Name: NO
State: Oslo
Locality: Oslo
Org Name: dave3610
```

```
Org Unit Name: dave3610
Common Name: server.dave3610.com
Mail:


# Edit apache config
ubuntu@first-instance:~$ sudo nvim /etc/apache2/sites-available/default-
ssl.conf

# Add lines
ServerName server.dave3610.com
SSLCertificateFile      /etc/ssl/certs/apache-selfsigned.crt
SSLCertificateKeyFile  /etc/ssl/private/apache-selfsigned.key
# Comment out snakeoil cert

# Reload apache
ubuntu@first-instance:~$ sudo a2ensite default-ssl.conf
ubuntu@first-instance:~$ sudo apache2ctl configtest
ubuntu@first-instance:~$ sudo systemctl reload apache2

# Run Squirrelmail config
ubuntu@first-instance:/var/www/html/squirrelmail$ ./configure
2. Server Settings
1. Domain: dave3610.com
S: Save data
Q: Quit

# Make folder for Squirrelmail data
ubuntu@first-instance:~$ sudo mkdir -p /var/local/squirrelmail/data/
ubuntu@first-instance:~$ sudo chown -R www-data:www-data
/var/local/squirrelmail/data

# Go to web interface
http://10.196.36.97/squirrelmail/src/login.php

# Login with sysadmin
sysadmin, password
```

SMTP-AUTH

- A secure (**SSL/TLS** protected) solution that will allow the employees to authenticate and send mail through the University SMTP server regardless of whether they are on Campus or somewhere else in the world.
    - In other words, the system must implement SMTP Authentication (**SMTP-AUTH**)
    - **NB!** You must use a self-signed SSL/TLS certificate generated by you and not Postfix

```
# Try to send mail with swaks with authentication
ubuntu@first-instance:~$ swaks  -tlso --to sysadmin@dave3610.com --from
'ubuntu@dave3610.com' --auth-user=sysadmin --auth-password=pagh3377 --server
127.0.0.1 --add-header 'X-Custom-Header: SMTP-AUTH Tested by Ismail' --body
'Testing SMTP-AUTH'
[…]
*** Host did not advertise authentication
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.

# Specify SASL plugin type that Postfix should use for authentication
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_sasl_type = dovecot'

# Specify the name of config file or rendevouz point for authentication
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_sasl_path = private/auth'

# Enable SASL authentication in the Postfix SMTP server
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_sasl_auth_enable = yes'

# Specify access restrictions for mail relay control in the context of RCPT TO
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_relay_restrictions =
permit_mynetworks, permit_sasl_authenticated, reject_unauth_destination'

# Check the Postfix config
ubuntu@first-instance:~$ postconf -n
smtpd_relay_restrictions = permit_mynetworks, permit_sasl_authenticated,
reject_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_path = private/auth
smtpd_sasl_type = dovecot
smtpd_tls_cert_file = /etc/ssl/certs/postfix.crt
smtpd_tls_key_file = /etc/ssl/private/postfix.key
smtpd_use_tls = yes

# Change the smtpd tls cert and key by generating my own
ubuntu@first-instance:~$ sudo openssl req -x509 -nodes -days 365 -newkey
rsa:2048 -keyout /etc/ssl/private/smtpd.key -out /etc/ssl/certs/smtpd.crt
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_tls_cert_file =
/etc/ssl/certs/smtpd.crt'
ubuntu@first-instance:~$ sudo postconf -e 'smtpd_tls_key_file =
/etc/ssl/private/smtpd.key'

# Check config again
ubuntu@first-instance:~$ postconf -n
smtpd_tls_cert_file = /etc/ssl/certs/smtpd.crt
smtpd_tls_key_file = /etc/ssl/private/smtpd.key

# Restart server and print log
```

```
ubuntu@first-instance:~$ sudo /etc/init.d/postfix restart; sudo tail -20
/var/log/mail.log

# Try to send mail with swaks with authentication again
ubuntu@first-instance:~$ swaks  -tlso --to sysadmin@dave3610.com --from
'ubuntu@dave3610.com' --auth-user=sysadmin --auth-password=pagh3377 --server
127.0.0.1 --add-header 'X-Custom-Header: SMTP-AUTH Tested by Ismail' --body
'Testing SMTP-AUTH'

# Check log to see if mail is delivered by postfix
ubuntu@first-instance:~$ sudo grep 'delivered to maildir' /var/log/mail.log
May  6 14:11:10 first-instance postfix/local[40060]: 3EBDBB8BE:
to=<sysadmin@dave3610.com>, relay=local, delay=0.28, delays=0.14/0.02/0/0.12,
dsn=2.0.0, status=sent (delivered to maildir)

# Read the email in MailDir
ubuntu@first-instance:~$ sudo bash -c 'cat /home/sysadmin/Maildir/new/*'
cat: '/home/sysadmin/Maildir/new/*': No such file or directory

# Got the mail in Squirrelmail and Thunderbird, so all is good. There are
probably some rights error on MailDir. Won't bother fixing it.
```

Spam and virus filter

- A mechanism that will detect and stop unsolicited bulk email (UBE) otherwise known as **SPAM**. Detected **SPAM** must be discarded and not quarantined in a special folder.
    - **Spamassassin** is the desired **Anti-SPAM** software for the University.
- A mechanism that will detect and stop **Viruses**. Detected **Virus** must be discarded and not quarantined in a special folder
    - **ClamAV** is the desired Anti-Virus software for the University.
    - **Amavisd-new** is the desired interface between mailer (MTA) and content checkers: virus scanners, and/or SpamAssassin.

```
# Install necessary packages
ubuntu@first-instance:~$ sudo apt install -qq -y amavisd-new spamassassin
clamav-daemon

# Install packages so amavis can scan compressed files
ubuntu@first-instance:~$ sudo apt install -qq -y arj cabextract cpio lhasa
nomarch pax rar unrar unzip zip

# Add the clamav user to the amavis group in order to have the appropriate
access to scan files
ubuntu@first-instance:~$ sudo adduser clamav amavis
Adding user `clamav' to group `amavis' ...
Adding user clamav to group amavis
Done.
ubuntu@first-instance:~$ sudo adduser amavis clamav
Adding user `amavis' to group `clamav' ...
Adding user amavis to group clamav
Done.

# Activate spam and virus detection in Amavisd-new by editing the content-
filter file
ubuntu@first-instance:~$ sudo nvim /etc/amavis/conf.d/15-content_filter_mode

@bypass_spam_checks_maps = (
   \%bypass_spam_checks, \@bypass_spam_checks_acl, \$bypass_spam_checks_re);

@bypass_virus_checks_maps = (
   \%bypass_virus_checks, \@bypass_virus_checks_acl, \$bypass_virus_checks_re);

# Start spamassassin and restart the amavis service
ubuntu@first-instance:~$ sudo systemctl enable spamassassin.service
Synchronizing state of spamassassin.service with SysV service script with
/lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable spamassassin
Created symlink
/etc/systemd/system/multi-user.target.wants/spamassassin.service →
/lib/systemd/system/spamassassin.service.
ubuntu@first-instance:~$ sudo systemctl start spamassassin.service
ubuntu@first-instance:~$ sudo systemctl restart amavis.service

# Configure Postfix to use the content_filter parameter with postconf command
ubuntu@first-instance:~$ sudo postconf -e 'content_filter = smtp-amavis:
[127.0.0.1]:10024'
```

```
# Check the Postfix config
ubuntu@first-instance:~$ sudo postconf -n
content_filter = smtp-amavis:[127.0.0.1]:10024

# Configure master.cf so Postfix know which program to send mail to for
filtering
# Copy the content from teacher, based on the Ubuntu mail filter docs
ubuntu@first-instance:~$ sudo nvim /etc/postfix/master.cf

smtp-amavis     unix    -       -       -       -       2       smtp
    -o smtp_data_done_timeout=1200
    -o smtp_send_xforward_command=yes
    -o disable_dns_lookups=yes
    -o max_use=20

127.0.0.1:10025 inet    n       -       -       -       -       smtpd
    -o content_filter=
    -o local_recipient_maps=
    -o relay_recipient_maps=
    -o smtpd_restriction_classes=
    -o smtpd_delay_reject=no
    -o smtpd_client_restrictions=permit_mynetworks,reject
    -o smtpd_helo_restrictions=
    -o smtpd_sender_restrictions=
    -o smtpd_recipient_restrictions=permit_mynetworks,reject
    -o smtpd_data_restrictions=reject_unauth_pipelining
    -o smtpd_end_of_data_restrictions=
    -o mynetworks=127.0.0.0/8
    -o smtpd_error_sleep_time=0
    -o smtpd_soft_error_limit=1001
    -o smtpd_hard_error_limit=1000
    -o smtpd_client_connection_count_limit=0
    -o smtpd_client_connection_rate_limit=0
    -o
receive_override_options=no_header_body_checks,no_unknown_recipient_checks,no_m
ilters

# Under pickup
pickup    unix  n       -       y       60      1       pickup
        -o content_filter=
        -o receive_override_options=no_header_body_checks

# Restart Postfix
ubuntu@first-instance:~$ sudo service postfix restart

# Download testfile from spamassassing to test if our server captures spam and
discards it
ubuntu@first-instance:~$ wget https://spamassassin.apache.org/gtube/gtube.txt
2024-05-07 07:34:22 (9.99 MB/s) - ‘gtube.txt’ saved [799/799]

# Test to send an email that contains spam
ubuntu@first-instance:~$ swaks -tlso --to sysadmin@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: SPAM
Tested by Ismail' --body gtube.txt

[…]
<~  250 2.0.0 Ok: queued as 6459AAABE
 ~> QUIT
<~  221 2.0.0 Bye
```

```
# Check the logs for discarded/blocked spam
ubuntu@first-instance:~$ sudo grep -i SPAM  /var/log/mail.log

# Fetch virus definitions for clamav
ubuntu@first-instance:~$ sudo freshclam


# Download a virus testfile from eicar.org
ubuntu@first-instance:~$ wget https://secure.eicar.org/eicar.com.txt
2024-05-07 07:38:56 (15.2 MB/s) - ‘eicar.com.txt’ saved [68/68]

# Send a mail with the virus testfile to see if it works
ubuntu@first-instance:~$ swaks -tlso --to sysadmin@dave3610.com --from
'ubuntu@dave3610.com' --server 127.0.0.1 --add-header 'X-Custom-Header: VIRUS
Tested by Ismail' --body eicar.com.txt

[…]
<~  250 2.0.0 Ok: queued as C4A75AFAB
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.

# Check the logs to see if the virus is discarded/blocked
ubuntu@first-instance:~$ sudo grep -i INFECTED  /var/log/mail.log

# Note: This did not actually work. I am not bothering troubleshooting it.
```

Multiple domains

- The ability to handle multiple email domains. It is very common these days for a single system to host many domains and handle email for them. The system must handle email for the domains: **dave3610.com**, **dave3610.no** and **dave3610.org**
    - **NB!** The email address **user1@dave3610.com** goes to the same mailbox as the address **user1@dave3610.no** and **user1@dave3610.org**

```
# Edit the main.cf config for Postfix to handle multiple domains
ubuntu@first-instance:~$ sudo nvim /etc/postfix/main.cf

virtual_alias_domains = dave3610.no dave3610.org
virtual_alias_maps = hash:/etc/postfix/virtual

# Set up virtual aliases
ubuntu@first-instance:~$ sudo nvim /etc/postfix/virtual

@dave3610.no      @dave3610.com
@dave3610.org     @dave3610.com

# Update Postfix
ubuntu@first-instance:~$ sudo postmap /etc/postfix/virtual
ubuntu@first-instance:~$ sudo systemctl reload postfix

# Test the configuration
ubuntu@first-instance:~$ swaks --to sysadmin@dave3610.no --from
'ubuntu@dave3610.com' --auth-user sysadmin --auth-password pagh3377 --server
127.0.0.1 --tls --add-header 'X-Custom-Header: Virtual Alias Tested by Ismail'
--body 'Testing virtual alias for dave3610.no'

[…]
<~  250 2.0.0 Ok: queued as 798FFBFF8
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.

ubuntu@first-instance:~$ swaks --to sysadmin@dave3610.org --from
'ubuntu@dave3610.com' --auth-user sysadmin --auth-password pagh3377 --server
127.0.0.1 --tls --add-header 'X-Custom-Header: Virtual Alias Tested by Ismail'
--body 'Testing virtual alias for dave3610.org'

[…]
<~  250 2.0.0 Ok: queued as E46CDC08D
 ~> QUIT
<~  221 2.0.0 Bye
=== Connection closed with remote host.

# Check the result
ubuntu@first-instance:~$ sudo grep 'sysadmin@dave3610.com' /var/log/mail.log
ubuntu@first-instance:~$ sudo grep 'sysadmin@dave3610.no' /var/log/mail.log

# Note: I get connection refused, but I won't bother troubleshooting this
```